PAGANI - SALSA ANALISI MATEMATICA I
(Zanichelli)

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Subject "Analisi 1"

INSIEMI A, B

$A = \{$ cane, gatto, pappagallo $\}$

$B = \{$ Lettere alfabeto italiano $\} \subseteq \{$ lettere alfabeto inglese $\}$

- $[0,1] = \{ x \in \mathbb{R} : 0 \leq x \leq 1 \}$ intervallo chiuso

  $\frac{1}{2} \in [0,1]$ $2 \in [0,1]$ $0,1 \in [0,1]$

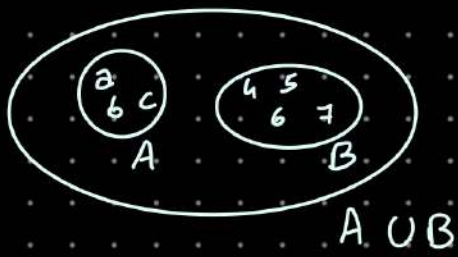
- $]0,1[\; = (0,1) = \{ x \in \mathbb{R} : 0 < x < 1 \}$ int. aperto

$[2,5[\; = \{ x \in \mathbb{R} : 2 \leq x < 5 \}$

UNIONE:

$A = \{a, b, c\}$ $B = \{4, 5, 6, 7\}$

$A \cup B = \{a, b, c, 4, 5, 6, 7\}$



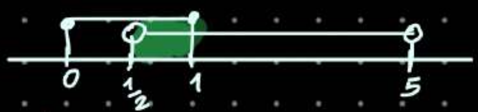
$A \cup B \overset{def}{:=} \{ x \in A$ oppure $x \in B \}$

ES: $[0,1] \cup [2,5)$

INTERSEZIONE

$A \cap B := \{ x \in A$ e $x \in B \}$

ES: $[0,1] \cap (\frac{1}{2}, 5) = \{ x \in \mathbb{R} : \frac{1}{2} < x \leq 1 \} = (\frac{1}{2}, 1]$



Intervalli a ∞

$(-\infty, a] = \{ x \in \mathbb{R} : x \leq a \}$

$(b, +\infty) = \; \sim \; \sim \; \sim$

∞ non è un numero (REALE)
non può essere MAI incluso
~~$(a, +\infty]$~~

Ricordare

- Gli insiemi si nominano MAIUSCOLO
  i suoi elementi (minuscoli)
  APPARTENGONO ($\in$) o non app. ($\notin$)
  ad esso

- $\subseteq$ inclusione
  (italiano incluso in inglese)

Es: $A = \{a, b, c\}$ $\quad B = \{b, c, d, e, f\}$

$A \cup B = \{a, b, c, d, e, f\}$ $\quad A \cap B = \{b, c\}$

FUNZIONI (A, B, f)

A Dominio $\quad$ B codominio / immagine

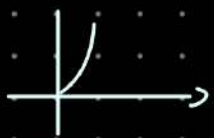
f è una relazione che per ogni $(\forall)$ $a \in A$ associa <u>uno e uno solo</u> $b \in B$

$f(\text{BLU}) = \#$
$f(\text{Giallo}) = ?$
$f(\text{VERDE}) = \sim$
$f(\text{ROSSO}) = \#$

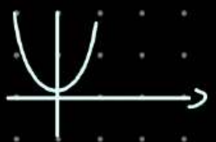
Ogni elemento del dominio, deve essere associato ad un elemento del codominio.

Def (f, A, B)

$f(x) = x^2 \qquad A = \{x \in \mathbb{R} : x \geq 0\} \quad B = \{x \in \mathbb{R}\}$



$f(x) = x^2 \qquad A = \{x \in \mathbb{R}\} \qquad B = \{x \in \mathbb{R}\}$



DEF: È INIETTIVA $f: A \to B$ $\quad$ f è una funzione con dominio A e codominio B

f si dice iniettiva se $\forall\ a_1, a_2 \in A \quad a_1 \neq a_2$
$f(a_1) \neq f(a_2)$

es: $f(x) = x^2 \qquad A[0,1] \quad B = \mathbb{R}$

$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$ $\qquad$ È iniettiva?

$x_1 \neq x_2 \overset{?}{\implies} x_1^2 \neq x_2^2$

$x_1 < x_2 \qquad x_1^2 - x_2^2 \neq 0$
$\qquad (x_1 - x_2)(x_1 + x_2)$

$0 \leq x_1 < x_2 \leq 1 \qquad x_1 - x_2 \neq 0 \qquad x_1 + x_2 > 0$

$x_1 < x_2 \implies x_1 - x_2 < 0$
$\qquad \implies (x_1 - x_2)(x_1 + x_2) < 0$
$0 \leq x_1 + x_2 \quad x_1 + x_2 > 0 \qquad x_1^2 - x_2^2 < 0 \implies x_1^2 < x_2^2$

(Proprietà $f(x) = x^2$ $x \in [0,1]$) vale che

$0 \leq x_1 < x_2 \implies f(x_1) < f(x_2)$

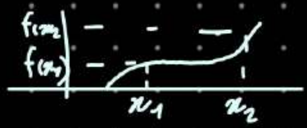
DEF: Funzione (strettamente) crescente

$f: A \to B$ $\quad A \subseteq \mathbb{R}$, $B \subseteq \mathbb{R}$

se $\forall\, x_1, x_2 \in A$ $\quad x_1 < x_2$

$\implies f(x_1) \leq f(x_2)$

$f(x_1) < f(x_2)$



$f(x) = x^2$ $\quad A = \mathbb{R}$ $\quad B = \mathbb{R}$

DEF IMMAGINE di f $\quad Im(f) \subseteq B$

$Im(f) = \{ b \in B,\ \exists\, a \in A : f(a) = b \}$

DEF f SURIETTIVA

$f: A \to B$ si dice suriettiva se

$Im(f) = B$ $\quad \forall b \in B \quad \exists\, a \in A : f(a) = b$

suriettiva
(ma non iniettiva)

$f(x) = \underset{\geq 0}{x^2}$ $\quad A = \mathbb{R}$ $\quad B = \mathbb{R}^+ = \{ x \in \mathbb{R} : x \geq 0 \}$

$x^2 \geq 0 \quad \vee \quad x^2 = 0 \iff x = 0$

f è suriettiva?

$y \in \mathbb{R}^+$ $\quad y \geq 0$ $\quad$ DEVO TROVARE $x \in \mathbb{R}$:

Summary

$f: A \to B$ $\quad$ funzione

f INIETTIVA $\quad \forall a_1, a_2 \in A \quad a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$

f SURIETTIVA $\quad \forall b \in B \quad \exists\, a \in A$ t.c. $f(a) = b$

$Im(f) \subseteq B$ $\quad Im(f) = \{ b \in B : \exists\, a \in A \quad b = f(a) \}$

Prodotto cartesiano di Due Insiemi

$A_1, A_2 \qquad A_1 \times A_2$

$$A_1 \times A_2 = \{(a,b) \quad a \in A_1,\ b \in A_2\}$$

$A_1 = \{a, b, c\}$

$A_2 = \{69, 420\}$

$$A_1 \times A_2 = \{(a,69), (b,69), (c,69), (a,420), (b,420), (c,420)\}$$

$$A_1 \times A_2 \times A_3 = \{(a,b,c),\quad a \in A_1, b \in A_2, c \in A_3\}$$

Grafico di f

$f: A \to B$

$G(f) \subseteq \underset{\text{dom}}{A} \times B^{\text{codom}}$

$G(f) := \{(a,b) \in A \times B : b = f(a)\}$

$A = \{1,2,3,4\}$

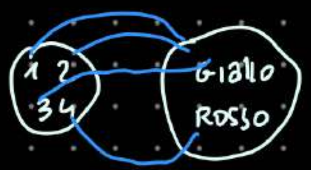
$B = \{\text{giallo}, \text{rosso}\}$

$f(1) = f(2) = f(3) = \text{giallo}$

$f(4) = \text{ROSSO}$

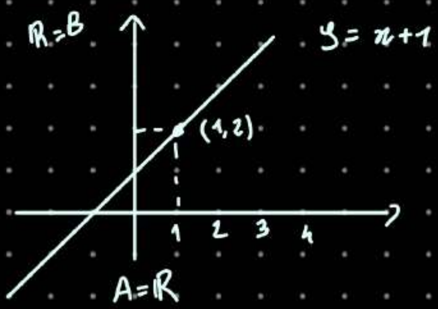
No iniett.

✓ suriett



$$A \times B = \{\underline{(1,g)}, \underline{(2,g)}, \underline{(3,g)}, (4,g), (1,r), (2,r), (3,r), \underline{(4,r)}\}$$

$G(f)$

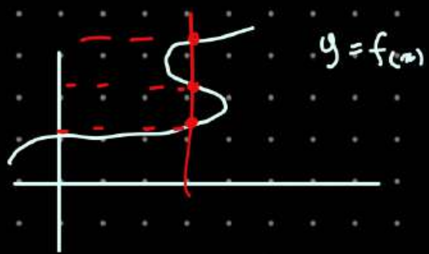


$(x,y) = (x, f(x))$

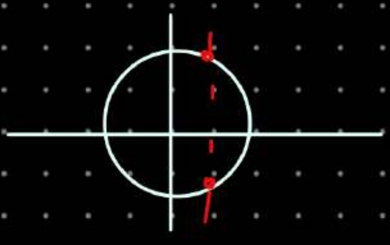
$f(1) = 2$

$A = \mathbb{R}$

$B = \mathbb{R}$ $\quad \to \quad \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$

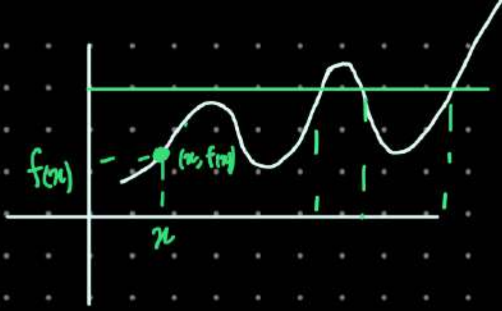


f associa a $x \in A$ UNO E UNO SOLO $y \in B$



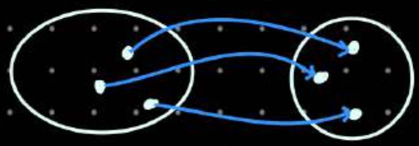
No funzione

↳ Luogo geometrico



DEF: $f : A \to B$ f iniettiva e suriettiva

f si dice Bigettiva (o nei libri inglesi 1 a 1)



Def: f. inversa

f bigettiva $f : A \to B$
g funz. inversa $g : B \to A$

$g(f_{(a)}) = a \quad \forall a \in A$

$f(g_{(b)}) = b \quad \forall b \in A$

es: $f(x) = x+1$

$2 \to 2+1$
$\pi \to \pi+1$

f è iniettiva? $x_1 \neq x_2$ e $f_{(x_1)} = f_{(x_2)}$

$-1 + x_1 + 1 = x_2 + 1 - 1$

$x_1 = x_2$

$g(x) = x-1$

$g(f_{(x)}) = x$

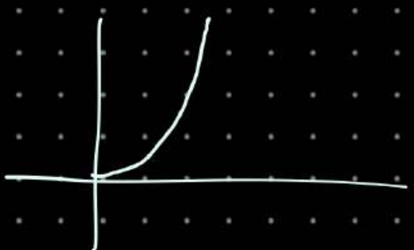
es

$f_{(x)} = x^2 \qquad g(y) = \sqrt{y}$

$\sqrt{f_{(x)}} = \sqrt{x^2} = g(F_{(x)}) \qquad \cancel{\sqrt{x^2} = x}$

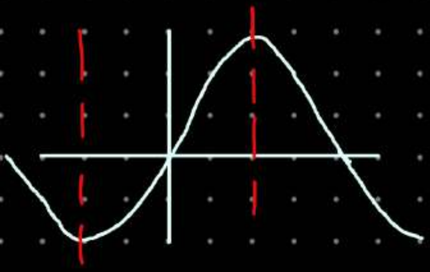
$\sqrt{(-3)^2} = \sqrt{9} = 3 \quad \neq -3$

$f(x) = x^2 \quad A = \{x \in \mathbb{R} : x \geq 0\} = B$



$$\sqrt{x^2} = |x| = \begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}$$

$\boxed{|x| = \pm x \quad \text{NO}}$



arcsin (x) è L'INVERSA di $f(x) = \sin(x)$ $\quad A = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$B = [-1, 1]$

Restrizione AD $A_1 \subseteq A$ $\quad f|_{A_1}$ $\quad f: A \to B$

$\sin(x)|_{[-\frac{\pi}{2}, \frac{\pi}{2}]}$

Numeri

$\mathbb{N}$ - Numeri naturali

$+$ $\quad + \; \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ $\quad n + m = m + n$

$\cdot$ $\quad \cdot$ $\quad n \cdot m = \underbrace{n + n + \dots + n}_{m \text{ volte}}$

$f: A \to \{1, 2, 3\}$

Bigettiva

$\# A = 3 \to$ la cardinalità di A è uguale a 3

Divisione con Resto

$a, b \in \mathbb{N}$ $\quad a \geq b$ $\quad a = b \cdot q + r$ $\quad q \in \mathbb{N}$ $\quad r \in \mathbb{N} \cup \{0\}$

$r < b$

$r = 0$ in qualche

☆ Divisione TRA Polinomi

Polinomio $\quad a(x) = a_0 + a_1 x + \dots a_n x^n$ $\quad a_m \neq 0$

$a_k \in \mathbb{R}$

$$= \sum_{k=0}^{m} a_k x^k$$

$m = \deg(a(x))$

GRado del Polin

$b(x) = \sum_{k=0}^{m} b_k x^k$ $\quad b_n \neq 0$ $\quad m \geq n$ $\quad q(x), r(x)$ polinomi

$\deg(r(x)) < \deg(r(x))$

$$a(x) = b(x)\, q(x) + r(x)$$

Ordinamento $\quad n, m \in \mathbb{N} \quad n \leq m$

$$n \leq m \;\; e \;\; m \leq n \iff n = m$$

Prop $\quad n \leq m \quad n, m \in \mathbb{N} \quad k \in \mathbb{N}$

$$n + k \leq m + k$$
$$nk \leq mk$$

Numeri INTeri

$\mathbb{Z} = \{ \ldots -3, -2, -1, 0, 1, 2 \ldots \}$

$n, m \in \mathbb{Z}$
$k \in \mathbb{Z}$

$$n \leq m$$
$$\Downarrow$$
$nk \leq km$ se $k \geq 0$

$nk \geq km$ se $k < 0$

$\mathbb{N}$	$\mathbb{Z}$
1	0
2	1
3	-1
4	2
5	-2
6	3

$f(2) = 1$
$f(4) = 2$
$f(6) = 3$

$f(2k) = k$

$$f(n) = \begin{cases} \frac{n}{2} & \text{se } n \text{ é pari} \\ -\frac{n-1}{2} & \text{se } n \text{ é dispari} \end{cases}$$

ES: $\quad f: \mathbb{N} \to \mathbb{Z}$ é iniettiva e suriettiva | Da DimosTrare |

DEF NUMERABILE

A é Numerabile se $\exists f: A \to \mathbb{N}$ Bigettiva

$\mathbb{Q}$ RAZIONALI

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\} \boxed{/\sim}$$

portate col minimo denominatore possibile

$x \in \mathbb{Q}$
$x \neq 0$
reciproco x^{-1} $\qquad x \cdot x^{-1} = x^{-1} \cdot x = 1$

$\frac{5}{8} \cdot \left(\frac{3}{5} - 5 + \frac{64}{128} \right)$ posso risolverla in un modo finito di passi

$\frac{1}{5} = 0{,}2$ $\quad$ ma $\quad \frac{1}{3} = 0{,}33333\ldots$

Per convenzione $\quad 0{,}\overline{9} = 1$

$f(x) = x^2$ é strettam. cresc. per $x \geq 0$

$0 \leq x_1 < x_2 \implies x_1^2 < x_2^2$

$1^2 < 2^2 = 4$ $\qquad (\sqrt{2})(\sqrt{2}) = (\sqrt{2})^2 = 2$

$1 < \sqrt{2} < 2$

posso provare a restringere il valore di $\sqrt{2}$
senza garanzia di finire mai

$\hookrightarrow$ numero decimale non periodico

$\sqrt{2} \neq \frac{p}{q}$ non esistono p, q $\quad p, q \in \mathbb{N}$

$\sqrt{2} + \sqrt{3}$ $\qquad \to$ Salta a pagine avanti

Funzioni

- Composizione di funzioni

sia data $f : x \to y$
$\qquad\qquad g : x' \to z$

se risulta che $Im(f) \subseteq Dom(g)$

possiamo definire $g \circ f : X \to Z$

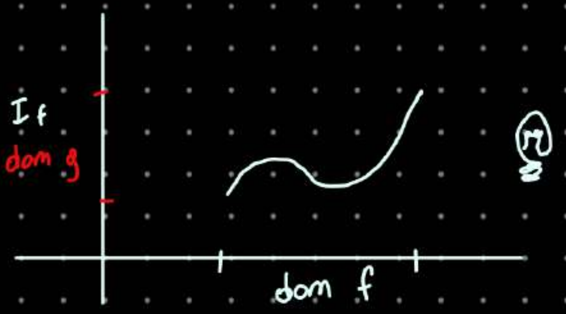
$$g \circ f(x) = g(f(x)) \qquad x \in X$$

Nota che $Im(f) \subseteq Dom(g)$

$$Im(g \circ f) = \{ \underbrace{g(f(x))}_{(g \circ f)(x)} : x \in X \} =$$

$$= \{ g(y) : y \in Im( ) \}$$

$$Im(g \circ f) = Im(g|_{Im(f)})$$



- SUCCESSIONI

$F: \mathbb{N} \to B$

$\forall n \in \mathbb{N}, \; f(n) \in B$

ESEMPIO

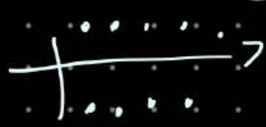
a) Lanci una moneta

$f(n) \in \{\text{testa, croce}\}$

b) $f(n) \in \{0, 1\}$

c) $f(n) = (-1)^n = \underbrace{(-1) \cdot \dots \cdot (-1)}_{n \text{ volte}}$

$= \begin{cases} 1 & \text{se } n \text{ è pari} \\ -1 & \text{se } n \text{ è dispari} \end{cases}$



d) Successione infinitesima

$f(n) \to 0$ se $n \to \infty$

$f(n) = \frac{1}{n}$

$\{f_n\}, \; f_n := f(n)$

Numeri reali

$D_{\mathbb{R}}(-\infty, +\infty)$

$\mathbb{R} = \{p, \alpha_1 \alpha_2 \dots \alpha_n \dots\} \quad p \in \mathbb{Z}$

$\alpha : \mathbb{N} \to \{0, 1, 2, 3, 4 \dots\}$

ESEMPI

$1 \quad (p = 1$ e $\alpha_1 = \alpha_2 = \dots \alpha_n \dots = 0$

$0,\overline{9} \quad (p = 0$ e $\alpha_n = 9 \;\; \forall n \in \mathbb{N})$

0,99999....

$2,835 \quad (p = 2$ e $\alpha_1 = 8, \alpha_2 = 3, \alpha_3 = 5 \;\; \alpha_n = 0 \;\; \forall \alpha \leq 4$

Teor $(\mathbb{Q} \subseteq \mathbb{R})$
(lemma) [proposizione < lemma < teorema]

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}$$

$$= \{ p_0, p_1, \overbrace{\cdots p_l}^{\text{Antiperiodo}} \overbrace{\overline{p_{l+1} \;\; p_r}}^{\text{periodo}} \}$$

$$p_0 \in \mathbb{Z}$$
$$p_i \in \{0, \dots, 9\}, \forall i \geq 1\}$$

- $2, 345\overline{19}$
- $1, \overline{91}$

DIM

1 $A \subseteq \mathbb{Q} \quad \wedge \quad \mathbb{Q} \subseteq A$

$$\implies A = \mathbb{Q}$$

2) $A \subseteq \mathbb{Q}$

Sia $a \in A$: $a =$

$$a \cdot 10^l = \underbrace{p_0 \; p_1 \cdots p_l}_{\mathbb{Z}}, \overline{p_{l+1} \cdots p_r}$$

$$a \cdot (10^l - 1) = p_0 p_1 \cdots p_l, \overline{(\dots)}$$

$$a \times 10^k = p_0 p_1 \cdots p_l p_{l+1} \cdots p_k , \overline{p_{l+1} \cdots p_k}$$

$$a \times 10^l = p_0 p_1 \cdots p_l , \overline{p_{l+1} \cdots p_k}$$

$$a \times (10^k - 10^l) = \underbrace{p_0 p_1 \cdots p_l p_{l+1} \cdots p_k - p_0 p_1 \cdots p_l}_{r \in \mathbb{Z}}$$

$$a = \frac{r}{10^k - 10^l}$$

$$\implies a \in \mathbb{Q}$$

3) Dobbiamo provare che $\boxed{\mathbb{Q} \subseteq A}$

Sia $a = \frac{p}{q}$, $\begin{matrix} p \in \mathbb{Z} \\ q \in \mathbb{N}. \end{matrix}$

(possiamo supporre che $p \geq 0$)

$p = p_0 q + m_1 \quad , \quad 0 \leq m_1 < q$

$39 \,|\, 4$ — $3{,}9700000$

Esempi
1 é raz
$0,\overline{83}$ é raz.
$\sqrt{2}$ non raz.

Lemma (di nuovo)

- $A \subseteq \mathbb{Q}$
- $\mathbb{Q} \subseteq A$

$a \in \mathbb{Q}, \ a \geq 0$

$$a = \frac{p}{q} \quad \begin{cases} p \in \mathbb{Z}, p \geq 0 \\ q \in \mathbb{N} \end{cases}$$

Algoritmo divisione

$$p = p_0 q + \overbrace{r_1}^{\text{RESTO}} \qquad 0 \leq r_1 \leq q$$

p, r_1, r_2 | q — p_0, p_1 ; $\{0, \dots 9\}$

$$\frac{10 \cdot r_1}{q} \leq \underbrace{\frac{10 \cdot q}{q}}_{=10}$$

ciò implica che

$0 \leq r_n < q$ $\qquad$ $r_n \in \{0 \dots, q-1\}$

$\Rightarrow \exists \ k \in \mathbb{N}$ t.c.

p, r_1, r_2, ..., r_ℓ, $r_{\ell+1}$, r_k | q — $p_0, \underline{p_1, \dots p_\ell}, \underline{p_{\ell+1} \dots p_k}$

$\mathbb{Q} \subseteq \mathbb{R}$

- consideriamo $a, b \in \mathbb{R}$

$a = p, \alpha_1 \alpha_2 \dots \alpha_n$

$b = q, \beta_1 \beta_2 \dots \beta_n$

Sono uguali se

$p = q \quad \wedge \quad \alpha = \beta$ → come successione decimale

Convenzione

- $0,\bar{9} \equiv 1$
- $p,\bar{9} = p+1$
- se $\alpha_k < 9$ allora

$= p, \alpha_1 \dots \alpha_k \bar{9} =$

$= p, \alpha_1 \dots (\alpha_k + 1) \bar{0} \dots$

ORDINAMENTO

1) per $a, b \in \mathbb{R}$ $\quad a, b \geq 0$

$\underset{(p,\alpha)}{a} \geq \underset{(q,\beta)}{b} \iff p > q \ \vee \ \begin{cases} p = q \\ \exists k \in \mathbb{N} \text{ t.c.} \\ \quad \alpha_n = \beta_n \ \forall n \in \{0 \dots k\} \end{cases}$

$\alpha_{k+1} \geq \beta_{\alpha+1}$

per esemio

- $3,8340 < 3,8341$
- $4,34 > 3,34$

2) se a è positivo e b è negativo $\Rightarrow a > b$

$(\forall a$ positivi, $a > 0)$

analogamente per a, b negativi

Somma e molt. in $\mathbb{R}$

$$1 - 0,\bar{9} = 0 \quad (?)$$

Idea: se $a \in \mathbb{R}$, possiamo incontrare una successione di reali $\{a^{(n)}\}$ t.c.

$$a^{(n)} \to a \qquad n \to \infty$$

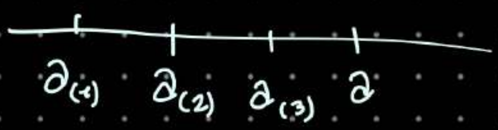
$$a+b = \lim_{n\to\infty} a^n + b^n$$

Dim

Sia $\begin{cases} a = p, \alpha_1 \dots \alpha_n \dots \in \mathbb{R} \\ a \geq 0 \end{cases}$

$$a_{(n)} := p, \alpha_1 \dots \alpha_n \bar{0}$$

$$a_{(n)} \leq a_{(n+1)} \leq a$$



$$|a - a_{(n)}| = a - a_{(n)} =$$

$$= p, \alpha_1 \dots \alpha_n \alpha_{n+1} \dots$$
$$- p, \alpha_1 \dots \alpha_n 0000 =$$
$$= 0,0 \dots 0 1 \overline{0000}$$

$$10^{-n} = \frac{1}{10^n} \to 0$$
$$n \to \infty$$

definiamo la somma di $a, b \in \mathbb{R}$ t.c.

$$a+b := \lim_{n\to\infty} \underbrace{\text{segno}(a)|a|_{(n)}}_{\substack{\downarrow \\ \text{sgn}(a)|a| \\ \parallel \\ a}} + \underbrace{\text{segno}(b)|b|_{(n)}}_{\substack{\downarrow \\ \text{sgn}(b)|b| \\ \parallel \\ b}}$$

$$a \cdot b := sgn(a) \cdot sgn(b) \cdot \lim_{n \to \infty} |a|_{(n)} \cdot |b|_{(n)}$$

$$1 - 0,\bar{9} =$$
$$= 1 + (-0,\bar{9})$$

$$1_{(n)} = 1$$
$$(0,\bar{9})_{(n)} = 0,\underbrace{99\ldots9}_{n\text{-volte}}$$

(n) quante volte voglio considerare la parte periodica

$$1_{(n)} - (0,9)_{(n)} = 0,0\ldots 1$$ → n.esimo risultato

Teorema: La Disuguaglianza Triangolare

qualsiasi $a, b \in \mathbb{R}$, soddisfano

$$|a+b| \leq |a| + |b|$$

Dim: notiamo come $\forall c \geq 0$, $|x| \leq c$

e' equivalente a ($\Leftrightarrow$)

$$\begin{cases} x \leq c & x \geq 0 \\ -x \leq c & x < 0 \end{cases}$$

$\Leftrightarrow$

$$-c \leq 0 \leq x \leq c \qquad \text{se } x \geq 0$$

$$-c \leq x < 0 \leq \qquad \text{se } x < 0$$

(*) $\Leftrightarrow$ $-c \leq x \leq c \quad \forall x$

Usiamo che $|a| = |a|$
$(|a| \leq |a|)$

poniamo $c = |a|$, $x = a$

si deduce da (*) che

$$-|a| \leq a \leq |a|$$

analogamente $-|b| \leq b \leq |b|$

Sommando

$x = a + b$
$c = |a| + |b|$

$$-(|a|+|b|) \leq a+b \leq |a|+|b|$$

$$|a+b| = |x| \leq c = |a| + |b|$$

Radianti e funzioni Trigonometriche

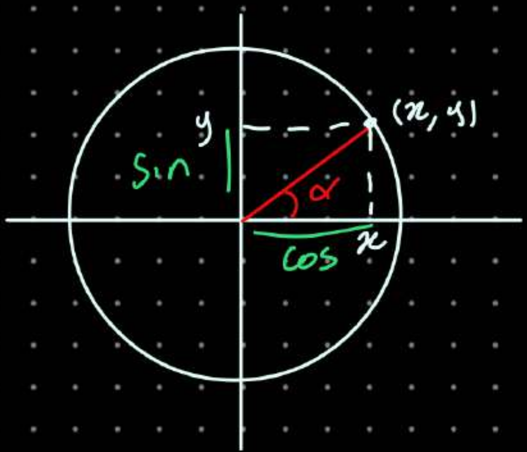
- Angoli in radianti

$r = 1$

$2\pi \rightarrow$ giro completo

$180° = \pi$ rad

$1° = \frac{\pi}{180}$ rad



$\cos(\alpha) = x$

$\sin(\alpha) = y$

$\tan(\alpha) = \frac{y}{x} = \frac{\sin(\alpha)}{\cos(\alpha)}$

$\cos\alpha \neq k\frac{\pi}{2}$ con $k \in \mathbb{Z}$

sin  cos  tan 

Fino a qui

$\sqrt{2} \notin \mathbb{Q}$

$(a+b)^2 = (a+b)(a+b) = a^2+2ab+b^2$

$(a+b)^3 = (a+b)^2(a+b) = a^3+3a^2b+3ab^2+b^3$

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

k che va da 0 a n

$\binom{n}{k}$ "n su k"

$n!$ Fattoriale

$n! = 1\cdot 2\cdot 3\cdot 4 \dots (n-1)\cdot n$

per convenzione $0! := 1$

$$\binom{n}{k} := \frac{n!}{k!\,(n-k)!}$$

Es $(a+b)^4 = 1a^4 \quad 3a^3b \quad a^2b^2 \quad ab^3 + 1b$

$\binom{4}{0} = \frac{4!}{0!\,(4-0)!} = 1 \qquad \binom{4}{4} = \frac{4!}{4!\,(4-4)!} = 1 \qquad \binom{4}{1} = \frac{4!}{3!} = 4$

$\frac{n!}{(n-1)!} = n \qquad \binom{n}{k} = \frac{n\,(n-1)\dots(n-k+1)}{k!}$

per non eseguire tutti questi calcoli laboriosi, si usa il Triangolo di Pascal - Tartaglia

n=
1) 1 1
2) 1 2 1
3) 1 3 3 1
4) 1 4 6 4 1
5) 1 5 10 10 5 1

Def n. pari / dispari

$n \in \mathbb{N}$ é pari $\overset{def}{\iff}$ n é divisibile per 2
$n = 2m \quad m \in \mathbb{N}$

n é dispari $\quad n = 2m-1 \quad m \in \mathbb{N}$

ES. $\mathbb{P} = \{ n \in \mathbb{N} \text{ t.c. } n \text{ é pari} \}$

$\exists \; f: \mathbb{P} \to \mathbb{N}$ Bigettiva

LEMMA a) n é pari $\Rightarrow$ n^2 é pari

$n = 2m \qquad n^2 = (2m)^2 = 4m^2 = 2(2 \cdot m^2)$

b) n é dispari $\Rightarrow$ n^2 é dispari

$n = 2m-1 \qquad n^2 = (2m-1)^2 = 4m^2 - 4m + 1$
$= 2(2m^2 - 2m) + 1$

Teorema $\sqrt{2} \notin \mathbb{Q}$

DIM Assurdo: Supponiamo $\exists \; p, q \in \mathbb{N}$ T.C. $\sqrt{2} = \frac{p}{q}$

ridotta ai MINIMI TERMINI

p, q PRIMI TRA LORO $\qquad MCD(p,q) = 1$

$(\sqrt{2})^2 = 2 \Rightarrow (\sqrt{2})^2 = \left(\frac{p}{q}\right)^2 = \quad 2 = \frac{p^2}{q^2}$

$\Rightarrow p^2 = 2q^2 \to$ É pari, per lo stesso motivo anche p^2

p^2 é pari $\Rightarrow$ p é PARI $\qquad p = 2m \quad m \in \mathbb{N}$

$2 = \frac{p^2}{q^2} \qquad 2 = \frac{(2m)^2}{q^2} = \frac{4m^2}{q^2}$

$1 = \frac{2m^2}{q^2} \Rightarrow q^2 = 2m^2$

$\Rightarrow q^2$ È PARI $\Rightarrow q$ é PARI

Dimostrato che q e p, pari, va in contraddizione con l'ipotesi che siano ridotte ai minimi Termini

---

$\# \mathbb{N} = \# \mathbb{Q}$

$\# \mathbb{R} > \# \mathbb{N}$

$\hookrightarrow$ non é insieme discreto ma continuo

1	0
2	1
3	-1
4	2
5	-2

$$f(n) = \begin{cases} \frac{n}{2} & n \text{ é pari} \\ -\left(\frac{n-1}{2}\right) & n \text{ é dispari} \end{cases}$$

$f: \mathbb{N} \to \mathbb{Z}$) é una SUCCESSIONE

una f che va Da $\mathbb{N}$ verso... $f(n) = f_n$

<u>Es.</u> f é Bigettiva

f é iniettiva $n, m \in \mathbb{N}$ $f(n) \neq f(m)$

se $n \in P$ (pari) $m \in D$ (disp.) $f(n) > 0$ $f(n) \leq 0$

se $n, m \in P$ $n \neq m$ MA $f(n) = f(m)$?

$n = 2n_1$ $\quad f(n) = \frac{2n_1}{2} = n_1 = m_1 = \frac{2m_1}{2} = f(m)$

$\#$

$m = 2m_1$ $\quad n_1 = m_1 \Rightarrow n = 2n_1 = m = 2m_1$

$n, m \in D$ Dir. stessa

f suriettiva $\forall k \in \mathbb{Z}$ $\exists n \in \mathbb{N} : f(n) = k$

$I_m(f) = \mathbb{Z}$

$k > 0$ $k = 18.000.256.821$ $f(2k) = \frac{2k}{2} = k$ $\quad k > 0$ $k \in \mathbb{Z}$ $n = 2k$ $f(n) = \frac{2k}{2} = k$

$k < 0$ $k = -\left(\frac{n-1}{2}\right) = f(n)$

$\Updownarrow$

$-k = \frac{n-1}{2}$

$\Updownarrow$

$-2k = n-1$

$\Updownarrow$

$n = 1-2k \in D$

$g: \mathbb{Z} \to \mathbb{N}$ $\quad g\,f(n) = n$

$f(g(m)) = m$

$$g(k) = \begin{cases} 2k & k > 0 \\ 1 & k = 0 \\ 1-2k & k < 0 \end{cases}$$

DEF min e Max

il = uno SOLO

$\emptyset \neq A \subseteq \mathbb{R}$ $m := \min(A)$ m é il minimo di A

① $m \leq a$ $\forall a \in A$

② $m \in A$

$M = \max A$

① $M \geq a$ $\forall a \in A$

② $M \in A$

min e Max sono proprietà di insiemi ordinati

$A = (0,1] = \{x \in \mathbb{R} : 0 < x \leq 1\}$

min A = ? min A = $\nexists$

se dicessi il min é $x = 0{,}000025$

Potrei dire, min é $0 < \frac{x}{2} < x$

A (5, 87) ⟶ 5 ... 87

fisso $x > 5$

il prossimo minimo sarà sempre

$5 < \frac{x+5}{2} < x$

↳ media aritm.

OSS $m = \min A$ m é unico

$m_1 = \min A$ $m_2 = \min A$ $m_1 \neq m_2$ (uno dei due é sbagliato)

$\begin{cases} m_1 \leq a \\ m_1 \in A \end{cases} \forall a \in A$ $\begin{cases} m_2 \leq a \\ m_2 \in A \end{cases} \forall a \in A$

$m_1 \leq a$ $m_2 \leq a$

$m_1 \leq m_2$ $\not\Leftrightarrow$ $m_2 \leq m_1$ $\Rightarrow m_1 = m_2$

$A \subseteq \mathbb{N} \Rightarrow \exists \min A$

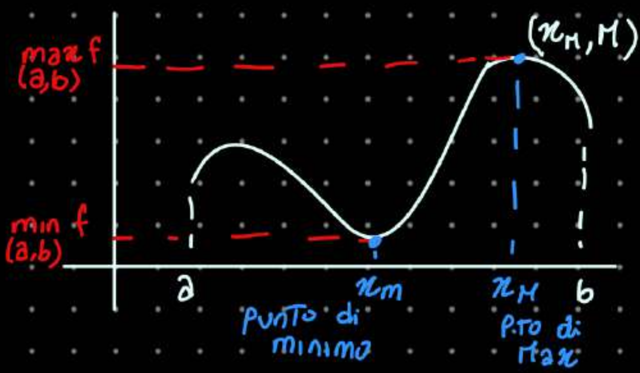
insieme discreto

0 1 2 3 4

DEF $f : A \to \mathbb{R}$ A qualsiasi

$m : \min_A f$) minimo di f sull'insieme A $\min_A f = \min Im|_f$

$M : \max_A f$ $\max_A f = \max Im|_f$



$A = (a,b)$
$B = \mathbb{R}$

$f(x_m) = m = \min_A f$

$f(x_M) = M = \max_A f$

ES f (cane) = 10 kg
f (gatto) = 5 kg
f (pesce) = 0,05 kg

cane é P.TO di MAX della funzione

$MCD(m,n) = (m,n)$

$= \max\{ d \in \mathbb{N} : d/m \text{ e } d/n \}$ divisori comuni m e n

d/m ↳ d divide m

$d/m :=$ d é divisore di m
d divide m ...

$m = p_1^{\alpha_1} \dots p_k^{\alpha_k} \quad n = p_1^{\beta_1} \dots p_k^{\beta_k}$ | fattori comuni con potenza minima

es $MCD(a,b) \quad a > b$

$a = bq + r$ resto, q quoziente $\quad a \leq r < b$

• Lemma

$MCD(a,b) = MCD(b,r)$

$A = \{ d \in \mathbb{N} : d/a \text{ e } d/b \} \overset{?}{=} B = \{ d \in \mathbb{N} \; d/b \text{ e } d/r \}$

(idea $A \subseteq B$ e $B \subseteq A$)

① $d \in A \quad d/a \text{ e } d/b \Leftrightarrow a = \underline{d}a' \quad b = \underline{db'} \quad a', b' \in \mathbb{N}$

$a = bq + r$

$\underline{da'} = \underline{db'}q + r \quad \Rightarrow 0 \leq r = da' - db'q = d(a' - b'q)$

$\Rightarrow r$ é multiplo di $d \Leftrightarrow d/z$

② $d/b \text{ e } d/r \quad \Rightarrow b = db' \quad r = dr' \quad b', r' \in \mathbb{N}$

$a = bq + r = db'q + dr' = d(b'q + r') \Rightarrow d/a$

Es $\underset{a}{MCD(475}, \underset{b}{55)} = MCD(\underset{b}{55}, \underset{r}{35})$

$475 = 55q + r$
$= 55 \cdot 8 + 35$ ($\frac{475}{55}$)

$55 \times 8 = 440$

$MCD(\underset{a_1}{55}, \underset{b_2}{35}) = MCD(35, 20)$

$55 = 35q + r = 35 \cdot 1 + 20$

$MCD(\underset{a_2}{35}, \underset{b_3}{20}) = MCD(20, 15)$

$MCD(\underset{a_4}{20}, \underset{b_4}{15}) = MCD(15, 5)$

$20 = 15 \cdot q + r = 15 \cdot q + 5$

$MCD(15, 5) = MCD(\sim \;)$

$15 = 5 \cdot q + r = 5 \cdot 3 = 0$

es: convertire in codice

es:

$$\begin{array}{r} 27\times \\ 36= \\ \hline 162 \\ 81 \\ \hline 972 \end{array}$$

$(a\cdot 10+b)(c\cdot 10+d)$

$(2\cdot 10\cdot 7)(3\cdot 10+6)=$

$= (2\cdot 3)10^2 + (2\cdot 6 + 7\cdot 3)10 + 7\cdot 6$

Le $\cdot$ richiedono complessita' computazionale
($\cdot 10$ sposta solo la virgola, non conta)

$x = x_1 \cdot 10 + x_0 \qquad y = y_1 \cdot 10 + y_0$

$xy = x_1 y_1 \cdot 10 + (x_1 y_0 + x_0 y_1)10 + x_0 y_0$

dovrò calcolare
1) $x_1 \cdot y_1$
2) $x_0 \cdot y_0$

$(x_1 + x_0)\cdot(y_1+y_0) = x_1 y_1 + \boxed{x_0 y_1 + x_1 y_0} + x_0 y_0$

$(x_1+x_0)\underset{3)}{\cdot}(y_1+y_0) - x_1 \cdot y_1 - x_0 y_0 = x_0 y_1 + x_1 y_1$

Algoritmo Karatsuba

Sono riuscito a eseguire 3 moltipl. invece che 4 -> Riducendo il tempo di calcolo

$0,\bar{3} = \frac{1}{3} = 0,333333\ldots$

$0,33 = \underbrace{[0,\bar{3}]_2}$ troncata alla seconda cifra decimale

$[0,\bar{3}]_n = 0,\underbrace{3\ldots3}_{n}$

fare $\sqrt{2}+\sqrt{3}$ correttamente sarebbe $[\sqrt{2}+\sqrt{3}]_n$

$[0,\bar{3}]_n = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} \cdots \frac{3}{10^n} = \sum_{k=1}^{n} \frac{3}{10^k}$

$0,\bar{3} = \frac{3}{10} + \frac{3}{100} + \ldots$ e così via

$? = \sum_{k=1}^{\infty} \frac{3}{10^k} ?$

$\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} \cdots \frac{3}{10^n} = 3\left[\frac{1}{10} + \frac{1}{100} \cdots \frac{1}{10^n}\right]$

$= \frac{3}{10}\left[1 + \frac{1}{10} + \left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)^3 + \left(\frac{1}{10}\right)^{n-1}\right]$

$= \frac{3}{10}\left[1 + q + q^2 + q^3 \cdots + q^{n-1}\right]$

PROGRESSIONE GEOMETRICA $\quad x_n = a_0 q^n \quad \forall n \in \mathbb{N}$

$\frac{x_{n+1}}{x_n} = q$ Ragione

Proposizione

PROP $1+q+q^2+\dots q^{n-1} = \dfrac{1-q^n}{1-q}$ $\quad q \neq 1 \quad \forall n \in \mathbb{N}$

si dimostra per induzione

proposiz. al passo 1

$P(1)$ é vera

$P(N) \Rightarrow P(N+1)$

$$1+2+3+4+\dots(n-1)+n = \underbrace{(n+1)+(n+1)+\dots+(n+1)}_{n/2 \text{ volte}}$$

$$\frac{n(n+1)}{2} = \sum_{k=1}^{n} k$$

$P(n)$ = la somma dei primi n naturali vale $\dfrac{n(n+1)}{2}$

$P(1)$ $\quad \sum_{n=1}^{1} k = 1 = \dfrac{n(n+1)}{2} = \dfrac{1(1+1)}{2} = 1$
$n=1$

Supponiamo che per un certo $N \in \mathbb{N}$ $\quad P(k)$ é vera $\quad k = 1,\dots,N$

$\Rightarrow P(N+1)$ é vera

$$\sum_{k=1}^{N} k = \frac{N(N+1)}{2} \Rightarrow \sum_{k=1}^{N+1} k = \frac{(N+1)(N+2)}{2}$$

vera per $N \in \mathbb{N}$

$$\sum_{k=1}^{N+1} k = (1+2+\dots(N-1)+N)\,N+1 = \sum_{k=1}^{N} k + (N+1)$$

$$= \frac{N(N+1)}{2} + N + 1 = \frac{N(N+1)+2(N+1)}{2}$$

$$= \frac{(N+1)(N+2)}{2}$$

ES $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ | DA FARE Per casa

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$P(1)$:

$$1 = \frac{1(1+1)(2+1)}{6}$$
$$1 = \frac{\cancel{6}}{\cancel{6}} \, 1$$

✓ è vera

supponiamo che $P(n)$ sia vera per $N \in \mathbb{N}$

$$\sum_{k=1}^{N} k^2 = \frac{N(N+1)(2N+1)}{6}$$

$\Rightarrow$ sarà vera anche per $N+1$

$$\sum_{k=1}^{N+1} k^2 = \frac{(N+1)(N+1+1)(2N+3)}{6} = \frac{(N+1)(N+2)(2N+3)}{6}$$

$$\sum_{k=1}^{N} k^2 + (N+1)$$

$$\frac{N(N+1)(2N+1)}{6} + (N+1)^2$$

$$\frac{N(N+1)(2N+1) + 6(N+1)^2}{6}$$

$$\cancel{\frac{2N^3 + 2N^2 + 2N + N + 6N^2 + 12N + 6}{6}}$$

$$\cancel{\frac{2N^3 + 8N^2 + 15N + 6}{6}}$$

$$(N+1)\left[N(2N+1) + 6(N+1)\right]$$

$$2N^2 + N + 6N + 6$$

$$2N^2 + 7N + 6 \rightarrow (N+2)(2N+3)$$

$$\frac{(N+1)(N+2)(2N+3)}{6} \checkmark$$

$$\sum_{n=0}^{n} q^k = \frac{1-q^{n+1}}{1-q} \qquad q \neq 1$$

$0,\bar{3} = \frac{3}{10} + \frac{3}{100} + \dots$ e così via

progressione geometrica

$a_n = a q^n$ $\qquad a \in \mathbb{R}$

$q \in \mathbb{R} \rightarrow$ RAGIONE della progressione geom

La Formula si verifica per intuizione

$n=0 \quad \sum_{k=0}^{0} q^k = q^0 = 1$ $\qquad \left. \frac{1-q^{k+1}}{1-q} \right|_{n=0} = \frac{1-q^{0+1}}{1-q} = 1$

↳ calcolando f in n=0

Supponendo che $P_{(N)}$ vale per, un certo $N \in \mathbb{N}$

$\overset{?}{\Longrightarrow}$ $P(N+1)$ è vera

$$\checkmark \sum_{k=0}^{N} q^k = \frac{1-q^{N+1}}{1-q} \overset{?}{\Longrightarrow} \sum_{k=0}^{N+1} q^k = \frac{1-q^{N+2}}{1-q} \quad ?$$

suppongo vera $\overset{??}{\longrightarrow}$

$$\sum_{k=0}^{N+1} q^k = (1+q+q^2 \dots + q^N) + q^{N+1} = \left(\sum_{k=0}^{N} q^k\right) + q^{N+1}$$

$$= \frac{1-q^{N+1}}{1-q} + q^{N+1} = \frac{1-q^{N+1} + (1-q)q^{N+1}}{1-q} =$$

$$= \frac{1-q^{N+1} + q^{N+1} - q^{N+2}}{1-q} =$$

$$= \frac{1-q^{N+2}}{1-q} \quad \checkmark$$

$$[0,3]_n = \frac{3}{10}\left(1+\frac{1}{10}+\dots+\frac{1}{10^{n-1}}\right) = \frac{3}{10}\left[\frac{1-\left(\frac{1}{10}\right)^n}{1-\frac{1}{10}}\right]$$

$$\sum_{k=0}^{n-1} q^k \qquad q = \frac{1}{10}$$

$$= \frac{3}{10}\,\frac{10}{9}\left[1-\left(\frac{1}{10}\right)^n\right] = \frac{1}{3}\left[1-\left(\frac{1}{10}\right)^n\right]$$

$$\frac{1}{3} > [0,\bar{3}]_n = \frac{1}{3} - [0,\bar{3}]_n = \frac{1}{3} - \frac{1}{3}\left[1-\left(\frac{1}{10}\right)^n\right] = \cancel{\frac{1}{3}} - \cancel{\frac{1}{3}} + \frac{1}{3}\left(\frac{1}{10}\right)^n$$

quante cifre decim devo scegliere

$$\frac{1}{3}\,\frac{1}{10^n} < \frac{1}{10^6} \qquad 10^n > \frac{10^6}{3}$$

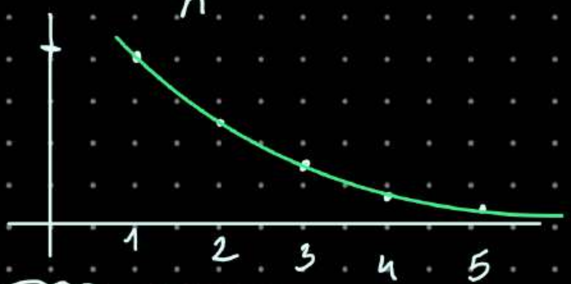
$$[0,\bar{3}]_n < [0,\bar{3}]_{n+1} < \frac{1}{3}$$

Ho modo per fare capire quante cifre dopo la virgola ci servono per avere errore minore di qualsiasi valore fissato

LIMITE di successione

Successione $f: \mathbb{N} \to \mathbb{R}$ $\quad f(n) = f_n$ "effe con n"

$a_n = \frac{1}{n} \qquad a_1 = 1 \quad a_2 = \frac{1}{2} \quad a_3 = \frac{1}{3}$



$\boxed{\max f = \max \mathrm{Im}(f)}$ A

$$\max_{n\in\mathbb{N}} a_n = \max_{n\in\mathbb{N}} \frac{1}{n} = \max_{\mathbb{N}} \frac{1}{n} = 1$$

il p.to di max è $x_M = 1$

$a_n = \frac{1}{n}$ è f decrescente

$\forall x,y \in D \quad x<y$

$f(x) \geq f(y)$

$n<m \quad \forall n,m \in \mathbb{N}$

$a_n \geq a_m$

$\Downarrow$

in realtà

$a_n > a_m$

(strettam. decresc.)

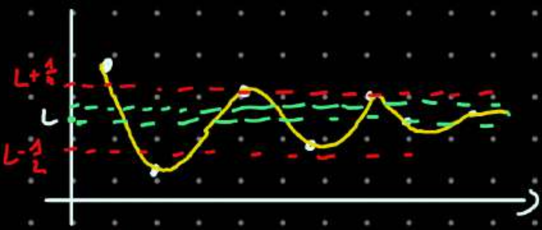
Def Limite di success.

$$\lim_{n \to +\infty} a_n = L \quad L \in \mathbb{R}$$

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ t.c. } \forall n > N \quad |a_n - L| < \varepsilon$$

$$-\varepsilon < a_n - L < \varepsilon$$

$$L - \varepsilon < a_n < L + \varepsilon$$



$\varepsilon = \frac{1}{n}$ ✗

come si calcola $\lim_{n \to +\infty} \frac{1}{n} = 0$

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ T.C. } \forall n > N \quad 0 - \varepsilon < \frac{1}{n} < 0 + \varepsilon$$

$$-\varepsilon < \frac{1}{n} < \varepsilon$$

Sempre $\frac{1}{n} > 0$

$-\varepsilon < 0$ $\Rightarrow -\varepsilon < \frac{1}{n}$ SEMPRE!

$\varepsilon = \frac{1}{4}$ $\quad \frac{1}{n} < \frac{1}{4} \iff n > 4$ $\boxed{N = 4}$ ⟶ ma io lo voglio $\forall$

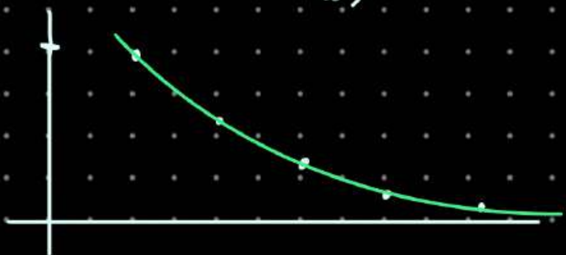
$\varepsilon > 0$ $\quad \frac{1}{n} < \varepsilon$ (da un certo punto in poi, per tutti gli $n > N$)

$\frac{1}{n} < \varepsilon \iff n > \frac{1}{\varepsilon}$

$\varepsilon = \frac{1}{100}$ $\quad n > \frac{1}{\frac{1}{100}} = 100 \; (= N)$

$\varepsilon = \frac{1}{10^9}$ $\quad n > \frac{1}{\frac{1}{10^9}} = 10^9 \ldots$

$\varepsilon = \sqrt{2}$



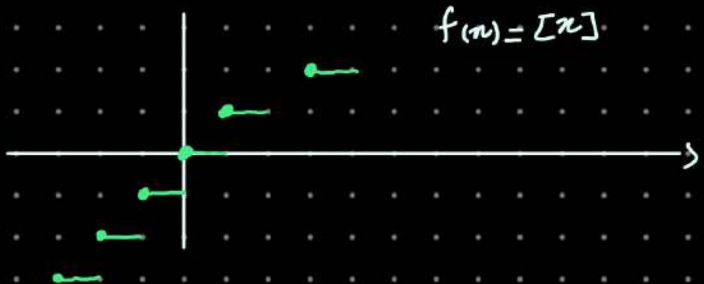
$[x]$ Parte intera di $x \in \mathbb{R}$

$$[x] \stackrel{def}{=} \{\max n \in \mathbb{Z},\ n \leq x\}$$

$x = 1{,}35\overline{8}$ $\quad y = 2{,}6458$ $\quad [1{,}358] = 1$

$[2{,}6458] = 2$

$f(x) = [x]$



$z = -\frac{1}{2} = -0{,}5$

$[-0{,}5] = -1$

Per casa, disegnare grafico

$\{x\} := x - [x]$ parte Frazionaria di x

$a_n = \frac{n}{n+1}$ $\quad a_1 = \frac{1}{2}$ $\quad a_2 = \frac{2}{3}$ $\quad a_3 = \frac{3}{4}$ $\quad a_4 = \frac{4}{5}$

- $\lim_{n \to +\infty} a_n \overset{?}{=} 1$ $\qquad 0 < \frac{n}{n+1} < 1 \iff n < n+1$

$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}$ t.c. $\forall n > N \quad 1 - \varepsilon < \frac{1}{n+1} < 1 + \varepsilon$

$\frac{n}{n+1} = \frac{n+1-1}{n+1} = \frac{n+1}{n+1} - \frac{1}{n+1} = 1 - \frac{1}{n+1}$

$\cancel{1} - \varepsilon < \cancel{1} - \frac{1}{n+1} < \cancel{1} + \varepsilon$

$-\varepsilon < -\frac{1}{n+1} < \varepsilon$ è gratis

resta da controllare $-\varepsilon < -\frac{1}{n+1}$

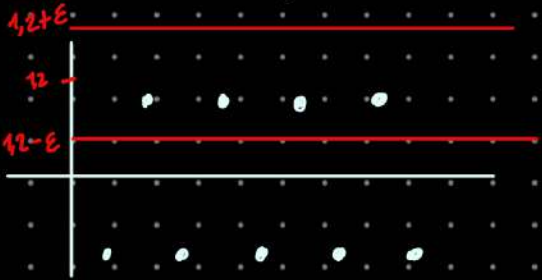
$\Updownarrow$

$\varepsilon > \frac{1}{n+1} \iff \frac{1}{n+1} < \varepsilon \iff n+1 > \frac{1}{\varepsilon}$

$\iff n > \frac{1}{\varepsilon} - 1$

$\boxed{n > \left[\frac{1}{\varepsilon} - 1\right] + 1}$

- $a_n = (-1)^n$ $\quad a_1 = -1$ $\quad a_2 = 1$ $\quad a_3 = -1$ $\quad a_4 = 1$



$\lim_{n \to +\infty} a_n = \underbrace{\lim}_{} (-1)^n = L$?

se il prof si dimentica di scrivere a cosa tende, è $+\infty$

$\lim_{n \to +\infty} (-1)^n =$ Non esiste

$\hookrightarrow \lim_{n \to +\infty} (-1)^n = 1{,}2$?

se fosse vero

$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}$ T.c. $\forall n > N \quad 1{,}2 - \varepsilon < (-1)^n < 1{,}2 + \varepsilon$

$\hookrightarrow$ però la "striscia" non comprende la parte < 0

TEOREMA

$$\lim_{n\to+\infty} a_n = L_1 \quad e \quad \lim_{n\to+\infty} a_n = L_2 \Rightarrow L_1 = L_2$$

Teorema algebra dei limiti

hp. $\lim_{a\to+\infty} a_n = L \qquad \lim_{n\to+\infty} b_n = M$

th

① $\lim_{n\to+\infty} (a_n+b_n) = L+M = \left(\lim_{n\to+\infty} a_n\right) + \left(\lim_{n\to+\infty} b_n\right)$

② $\lim_{n\to+\infty} (a_n \cdot b_n) = L \cdot M$

③ $\lim_{n\to+\infty} \left(\frac{a_n}{b_n}\right) = \frac{L}{M}$ se $M \neq 0$

$\lim_{n\to+\infty} c = c$

① $\forall \varepsilon > 0 \ \exists N \in \mathbb{N} : \forall n > N \ |(a_n+b_n) - (L+M| < \varepsilon$ tesi

---

limit $\{a_n\}$ $\hookrightarrow \forall \varepsilon > 0 \ \exists N \in \mathbb{N} : \forall n > N \ |a_n - L| < \varepsilon$

$\lim_{n\to+\infty} a_n = L \in \mathbb{R}$

$a_n = \frac{1}{n} \quad n \in \mathbb{N} \qquad b_n \begin{cases} e^{e^{e^n}} & 1 \leq n \leq 10^{100} \\ \frac{1}{n} & n > 10^{100} \end{cases} \qquad \lim_{n\to+\infty} b_n = 0$

Proprietà è vera definitivamente se $\exists M \in \mathbb{N}$ t.c. è vera da M in poi

$\lim_{n\to+\infty} \sqrt[n]{a} = 1 \quad a > 0 \qquad \lim_{n\to+\infty} \left(\sqrt[n]{20} + \frac{1}{n^2}\right) \quad a_n = \underbrace{\sqrt[n]{20}}_{b_n} + \underbrace{\frac{1}{n^2}}_{c_n}$

$c_n = \frac{1}{n^2} = \underbrace{\frac{1}{n}}_{d_n} \cdot \underbrace{\frac{1}{n}}_{e_n} \to \lim \frac{1}{n}\frac{1}{n} = 0 \cdot 0 = 0$

Teorema algebra dei Limiti

hp. $\lim_{n\to+\infty} a_n = L \qquad \lim_{n\to+\infty} b_n = M$

th
① $\lim_{n\to+\infty}(a_n+b_n) = L+M = \left(\lim_{n\to+\infty} a_n\right) + \left(\lim_{n\to+\infty} b_n\right)$

② $\lim_{n\to+\infty}(a_n \cdot b_n) = L \cdot M$

③ $\lim_{n\to+\infty}\left(\frac{a_n}{b_n}\right) = \frac{L}{M}$ se $M \neq 0$

$\lim_{n\to+\infty} c = c$

① $\forall \varepsilon > 0 \ \exists N \in \mathbb{N} : \forall n > N \ |(a_n+b_n) - (L+M)| < \varepsilon$ tesi

$\left.\begin{array}{l}\forall \varepsilon > 0 \ \exists N_1 \in \mathbb{N} : \forall n > N_1 \ |a_n - L| < \varepsilon \\ \forall \varepsilon > 0 \ \exists N_2 \in \mathbb{N} : \forall n > N_2 \ |b_n - M| < \varepsilon\end{array}\right\}$? ipotesi

$|a_n+b_n-L-M| = |a_n-L+b_n-M| \leq \underbrace{|a_n-L|}_{\hat{}\ \frac{\varepsilon}{2}\ \text{se } n>N_1} + \underbrace{|b_n-M|}_{\hat{}\ \frac{\varepsilon}{2}\ n>N_2} < \varepsilon$ piccolo?

$|x+y| \leq |x|+|y| \quad \forall x,y \in \mathbb{R}$

Se $n > \max\{N_1, N_2\} = N$

$\lim a_n = L \qquad \lim b_n = M \Rightarrow \lim (a_n+b_n) = L+M$

$\overset{?}{\Leftarrow}$ NO

$a_n = (-1)^n \qquad b_n (-1)^{n+1}$

$-1, 1, -1, 1, \dots \qquad 1, -1, 1, -1, \dots$

$a_n + b_n = 0$ (due succ le quali somma $\neq$ limiti)

quando sommiamo successioni che non convergono, ogni risultato è possibile

$a_n = (-1)^n \quad b_n = (-1)^n \Rightarrow$ no limite

$a_n \cdot b_n = (-1)^{2n} = 1$

$a_n = \frac{1}{n} \qquad b_n = \begin{cases} 0 & 1 \leq n \leq 100 \\ \sqrt[n]{5} & n > 100 \end{cases} \qquad b_n \to 1$

$\downarrow$
0

$C_n = \frac{a_n}{b_n} \quad n > 0$

Teor (Permanenza segno)

$$\lim_{n\to+\infty} b_n = M \quad \text{(numero)}$$

se $M \neq 0 \Rightarrow b_n$ é definitiv. $\neq 0$

$M > 0 \Rightarrow b_n$ " " > 0
$\exists\, K \in \mathbb{N},\ b_n > 0 \quad \forall n > K$

$M < 0 \Rightarrow \exists\, K \in \mathbb{N},\ b_n < 0 \quad \forall n > K$

"controesempio" $\lim b_n = 0$??

- $b_n = \frac{(-1)^n}{n}$ $\quad -1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}$

- $b_n = \begin{cases} -1 & 1 \leq n \leq 1000 \\ 1 & n > 1000 \end{cases}$ $\quad \lim b_n = 1 > 0$ non é vero che $b_n > 0$ "sempre"

Dimostrazione

hp $(M>0)$ $\quad \forall \varepsilon > 0 \quad \exists N \in \mathbb{N} : \forall n > 0$ $\quad$ Defin. di limite → succes

$0 < \frac{M}{2} = M - \frac{M}{2} = \underset{hp}{M} - \varepsilon < b_n < M + \varepsilon \quad \forall n > N$

$\varepsilon = \frac{M}{2}$

Allora

- $\lim_{n\to+\infty} \sin\left(\frac{1}{n}\right) = 0 \qquad \lim_{n\to+\infty} \frac{1}{n^k} = 0 \quad \forall k \in \mathbb{N}$

- $\lim_{n\to+\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$

Teorema confronto / sul segno

importante guardare i segni

$a_n \geq 0 \quad \forall n \in \mathbb{N}, \quad \lim_{n\to+\infty} a_n = L \Rightarrow L \geq 0$

$a_n = \frac{1}{n} > 0 \qquad \lim_{n\to+\infty} a_n = L \geq 0$

Dim per assurdo (no import)

Suppongo $\underset{\neq \text{tesi } (L \geq 0)}{L < 0}$ $\quad \lim_{n\to+\infty} a_n = L < 0$

$\exists\, N \in \mathbb{N}$ t.c. $a_n < 0 \quad \forall n > N$

$L + \varepsilon < a_n < L + \varepsilon$

$\emptyset = A \subseteq \mathbb{R}$ **è limitato**

$\exists \, k \in \mathbb{R} : |x| \leq k \quad \forall x \in A$

$-k \leq x \leq k$

Prendo un numero (k)
Tutti i p.ti (x) del mio insieme (A) sono compresi tra $+k$ e $-k$

a) A è **limit. superiormente**

$\exists \, k \in \mathbb{R} : x \leq k \quad \forall x \in A$

b) $A \subseteq \mathbb{R}$ **È lim. inferiormente** $(\exists \, k \in \mathbb{R} \quad k \leq x \quad \forall x \in A)$

$A_1 = (-20, 60)$ Limitato

$A_2 = \mathbb{N}$ Non Limitato sup.
-> Limit. inferiorm

DEF $\alpha \in \mathbb{R}$ è un **MAGGIORANTE** per A se $x \leq \alpha \quad \forall x \in A$

Proprietà insiemi lim. sup

DEF $\beta \in \mathbb{R}$ è un **MINORANTE** per A se $\beta \leq x \quad \forall x \in A$

$A = \{a_n\} \subseteq \mathbb{R}$

$a_n = \frac{1}{n}$ 

LIMIT.

$B = \{(-1)^n\} \subseteq \mathbb{R}$ 

È limitato

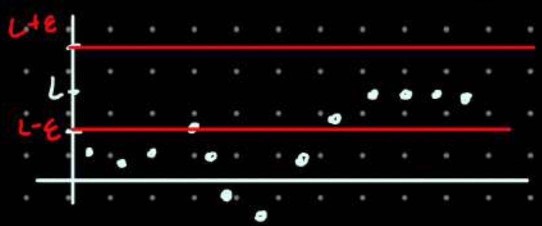
$a_n = (-1)^n$ non ha limite $\qquad (-1)^n$ è limitato

a_n limitato $\cancel{\Longrightarrow}$ a_n ammette limite

$|a_n| \leq k \quad \Longleftarrow \quad \lim_{n \to +\infty} a_n = L$

TEOREMA a_n ammette limite L $\Longrightarrow$ a_n è limitato

$\lim a_n = L \qquad\qquad \exists \, k \in \mathbb{R} \quad |a_n| \leq k$



$\varepsilon = 1 \qquad (L-1, L+1) \qquad L-1 < a_n < L+1 \quad \forall n > N$

$a_1, a_2, \ldots, a_{N-1}, a_N$

$m = \min_{1 \leq k \leq N} a_k$ $\qquad$ $M = \max_{1 \leq k \leq N} a_k$

$\emptyset \neq A \subseteq \mathbb{R}$ $\qquad$ $\min_{n \in N} \frac{1}{n} = N.E$

Teorema

$\emptyset \neq A \subseteq \mathbb{R}$ $\quad$ A Lim. Sup.

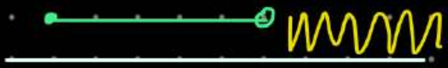
$\Rightarrow \exists \Lambda \in \mathbb{R}$ estr. Sup.

Lambda grande

ESTREMO SUPERIORE

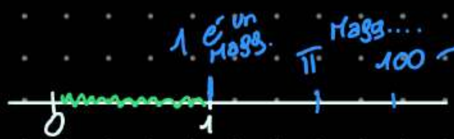
$\Lambda = \text{Sup}\, A$ $\qquad$ (Λ è l'estremo sup. di A)

Λ E' il minimo dei maggioranti di A



k è un maggiorante

Es • $A = (0,1)$ $\quad$ Max A = NE $\rightarrow$ Sup A = 1



• $B = (0,5) \cup \{6\}$ $\qquad$ Max B = 6 = Sup B

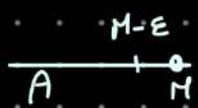
ma come si trova l'estr. Sup?

Λ è il min dei maggioranti

1. Λ è maggiorante
2. Λ è il minimo dei maggioranti

1. $x \leq \Lambda \quad \forall x \in A$
2. $\forall \varepsilon > 0 \quad \exists a \in A : a > \Lambda - \varepsilon$

1. $x \leq M \quad \forall x \in A$
2. $M \in A$



$\Lambda = \max A \Rightarrow \Lambda = \text{Sup}\, A$

ES

$A = \left\{ \frac{n}{n+1},\ n \in \mathbb{N} \right\}$

$0 < \frac{n}{n+1} < 1$

Max A = NE Sup $A \overset{?}{=} 1$

① $\forall x \in A$ $1 = \Lambda$ $1 - \Lambda \geq x$ $\frac{n}{n+1} \leq 1 \iff n \leq n+1$ Vera

"uno" è un maggior.

② $\forall \varepsilon > 0 \quad \exists\, x \in A$ T.C. $x > 1 - \varepsilon$

Devo provare n T.c. $\frac{n}{n+1} > 1 - \varepsilon \rightarrow \frac{n+1-1}{n+1} > 1 - \varepsilon$

$\cancel{1} - \frac{1}{n+1} > \cancel{1} - \varepsilon$

$\frac{1}{n+1} < \varepsilon$

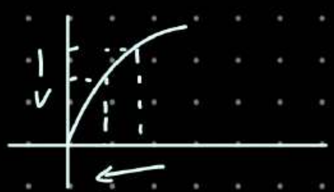
$n + 1 > \frac{1}{\varepsilon}$

$n > \frac{1}{\varepsilon} - 1 \qquad n \in \mathbb{N} \qquad n > \left[\frac{1}{\varepsilon} - 1\right] + 1$

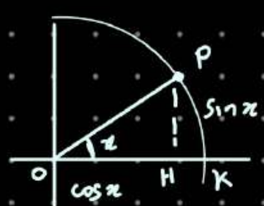
$\lim_{n \to +\infty} \sin\left(\frac{1}{n}\right) = 0$

$0 < \frac{1}{n} \leq 1$

$0 < \sin\left(\frac{1}{n}\right) \leq 1 \quad \forall n \in \mathbb{N}$



$\lim \left(\frac{1}{n}\right) = 0$ congett. $\lim \left(\frac{1}{n}\right) = 0$



$\overline{OP} = 1$

$\overline{OH} = \cos x$

$\overline{PH} = \sin x$

$A(O\hat{K}P) = \frac{\overline{OK} \cdot \overline{HP}}{2} = \frac{\sin x}{2}$

$\wedge$

$\frac{A(O\hat{K}P)}{A\,(\text{cerchio})} = \frac{x}{2\pi}$

$\frac{A(O\hat{K}P)}{\pi} = \frac{x}{2\pi}$

$A(O\hat{K}P) = \frac{\sin x}{2} \leq \frac{x}{2} = A(O\hat{K}P)$

★ $\sin x \leq x \quad \forall x \in \left(0, \frac{\pi}{2}\right)$

$|\sin x| \leq |x| \quad \forall x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$a_n = \quad 0 \leq \sin\left(\frac{1}{n}\right) \leq \frac{1}{n} = b_n \qquad x = \frac{1}{n} \qquad 0 < \frac{1}{n} \leq \frac{\pi}{2}$

TEOREMA DEL CONFRONTO (Dei carabinieri)

↳ Pinocchio non passa dalla porta e i carab. lo spingono

$\lim a_n \to L$

$\lim b_n \to L$

$a_n \leq c_n \leq b_n \quad \forall n \in \mathbb{N}$

il limite di c_n $\exists$

$\lim_{n \to +\infty} c_n = L$

$\forall \varepsilon > 0 \quad L - \varepsilon < b_n < L + \varepsilon \quad \forall n > N_1$

$\forall \varepsilon > 0 \quad L - \varepsilon < a_n < L + \varepsilon \quad \forall n > N_2$

$L - \varepsilon < a_n \leq \underline{c_n} \leq b_n < L + \varepsilon \quad \forall n > \max\{N_1, N_2\}$

$\lim_{n \to +\infty} \cos\left(\frac{1}{n}\right)$

$\sin^2 x + \cos^2 x = 1 \quad \forall x \in \mathbb{R}$

$\cos x = \sqrt{1 - \sin^2 x} \quad x \in \left(0, \frac{\pi}{2}\right]$

$\cos\left(\frac{1}{n}\right) = \sqrt{1 - \sin^2 \frac{1}{n}} = \sqrt{1 - \underset{0}{\sin\left(\frac{1}{n}\right)}\underset{0}{\sin\left(\frac{1}{n}\right)}} = 1$

$\lim_{n \to +\infty} \sqrt{a_n} \quad$ e $\lim a_n = L \quad n \geq 0$

$\Rightarrow \lim \sqrt{a_n} = \sqrt{L}$

$n \mapsto a_n \mapsto \sqrt{a_n} \qquad f(a_n) \quad f(x) = \sqrt{x}$

$\lim f(a_n) \overset{??}{=} f(\lim a_n)$

hp $\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} : \forall n > N \quad |a_n - L| < \varepsilon$

Th $\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} : \forall n > N \quad |\sqrt{a_n} - \sqrt{L}| < \varepsilon$

dim

$(L > 0) \qquad \sqrt{a_n} - \sqrt{L} = (\sqrt{a_n} - \sqrt{L}) \frac{(\sqrt{a_n} + \sqrt{L})}{(\sqrt{a_n} + \sqrt{L})} = \frac{a_n - L}{\sqrt{a_n} + \sqrt{L}}$

$|\sqrt{a_n} - \sqrt{L}| = \frac{|a_n - L|}{\sqrt{a_n} + \sqrt{L}}$

$$\left|\sqrt{a_n}-\sqrt{L}\right| \overset{\text{" stima"}}{=} \frac{|a_n-L|}{\sqrt{a_n}+\sqrt{L}} \leq \frac{|a_n-L|}{\sqrt{L}} < \frac{\varepsilon}{\sqrt{L}} \quad \text{se } n>N$$

10/10/2025

$$(a^2+2ab+b^2)=(a+b)^2$$

- $x^2+2x-5=0$

$x^2+2x+1-6$

$(x+1)^2=6$

$x+1=\pm 6$

$x=-1\pm\sqrt{6}$

- $x^2+2x+5=0$

$x^2+2x+1+4=0$

$(x+1)^2 \neq -4$

$x+1=\pm\sqrt{-4}$ No soluzioni Reali

$x=-1\pm\sqrt{-4}$

aggiungiamo un nuovo simbolo per "superare" questo Problema

$i^2 := -1 \qquad i \notin \mathbb{R}$ unità immaginaria

$\sqrt{-4}=\sqrt{-1\cdot 4}=i2$

$x=-1\pm i2$

$$x=p\pm iq \quad \begin{cases} \Delta=b^2-4ac \\ \Delta\geq 0 \quad q=0 \\ \Delta<0 \quad q\neq 0 \end{cases}$$

DEF NUMERI COMPLESSI

$$\mathbb{C}=\{z=a+ib \quad a,b\in\mathbb{R}\} \qquad z \longleftrightarrow (a,b)$$

a = parte reale di z
b = coeff. parte immag

OPERAZIONI

$z=a+ib$

$w=c+id$

$$(a,b)+(c,d)=(a+c,\, b+d)$$

$z+w=(a+c)+i(b+d)$

OSS se $b,d=0$

$z+w=a+c+i(0+0)$
$=a+c$

Piano di Argand e Gauss

$z = 1 + i0$
$w = 0 + i1$

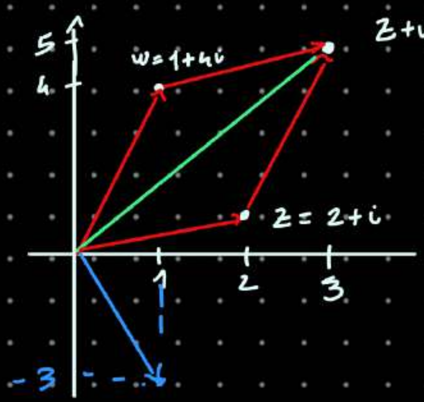
Nei $\mathbb{C}$ Vale la:

- $\exists$ numero neutro

$0_{\mathbb{C}} = 0 + i0$

- Propr. commutativa

$z + w = w + z$



$\underline{z - w} = (2-1) + i(1-4)$

- P. Associativa

$(z + w) + t = z + (w + t)$
$\forall\, z, w, t \in \mathbb{C}$

$z - w$

$z -$ "opposto di w"

$z + a + ib$
$w = c + id$

$-w = -c - id$

$w + (-w) = c + id + (-c - id)$
$\overset{!}{=} 0 + i0$

- Prodotto di $\mathbb{C}$

$$\begin{cases} (a,b) \underset{\mathbb{C}}{+} (c,d) = (a \overset{T}{+} c, b+d) \quad + \text{ tra numeri} \\ (a,b) \underset{\mathbb{C}}{\cdot} (c,d) = (\,?\,,\,?\,) \end{cases}$$

$z = a + ib$

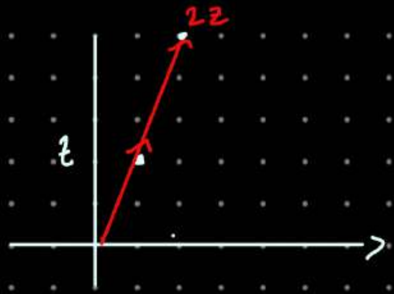
$w = c + id$

$$(a + ib) \cdot (c + id) = ac + iad + ibc + \underbrace{ib\,id}_{i^2}$$
$$= ac - bd + i(ad + bc)$$
$$= (ac - bd) + i(ad + bc)$$

$z = 3 + 0i$
$w = 2 + 0i$
$z \cdot w = (3 + 0 \cdot 2)(2 + 0 \cdot 6)$

$= 6 + 0 + i(0 + 0) = 6 + i0$

Posso "IDENTIFICARE" $\mathbb{R}$ con $\mathbb{R} = \{z \in \mathbb{C} : z = (a, 0)\}$
$a \in \mathbb{R}$



$w = c + id$
$w = 2 + 0i$

$z = 1 + 2i$

$wz = 2 + 4i$

ottengo un numero nella stessa direzione e stesso verso se $c > 0$
opposto se $c < 0$

- Lunghezza $z = a + ib$ È la DISTANZA di (a,b) da $(0,0)$

MODULO di $z \in \mathbb{C}$

$\|z\| := \sqrt{a^2+b^2}$

$\sqrt{a^2+b^2}$

$\|z\| > 0 \qquad \|z\| = 0 \iff z = 0 + i0 \qquad \forall z \in \mathbb{C}$

$\|\lambda z\| = |\lambda| \, \|z\| \qquad \lambda \in \mathbb{R} \quad z \in \mathbb{C}$

$\lambda z = \lambda a + i \lambda b \qquad \|\lambda z\| = \sqrt{(\lambda a)^2 + (\lambda b)^2} = \sqrt{\lambda^2 a^2 + \lambda^2 b^2}$

$= \sqrt{\lambda^2} \, \sqrt{a^2+b^2}$

$= \underbrace{|\lambda|}_{\text{numero} \in \mathbb{R}} \sqrt{a^2+b^2}$

$|2 + i3| = \| 2^2 + i(3)^2 \| = \sqrt{4+9}$

$\neq \sqrt{4-9}$

non $2^2 + (i3)^2$

$z \cdot w = w \cdot z$

$(z+w)t = zt + wt$

$\|z\| = \sqrt{a^2+b^2} \qquad z \in \mathbb{C}, \; z = a + ib$

$\|x\| = \sqrt{x \cdot x} \qquad x \in \mathbb{R}$

$\|z\| \neq \sqrt{z \cdot z} \qquad z^2 = z \cdot z = (a+ib)(a+ib) = a^2 - b^2 + i(ab+ba)$

$= a^2 - b^2 + 2iab$

$i \cdot i = (0+i)(0+i) = -1 + i0 \qquad (0,1) \cdot (0,1) = (-1,0)$

Unità neutra molt.

$z \cdot \mathbb{1}_{\mathbb{C}} = \mathbb{1}_{\mathbb{C}} \cdot z = z$

$z = a + ib \qquad (a+ib)(c+id) = a + b \qquad c = 1$
$\mathbb{1}_{\mathbb{C}} = c + id \qquad ac - bd + i(ad+bc) = a + ib \qquad d = 0$

$\mathbb{1}_{\mathbb{C}} = 1 + i0 = (1,0)$

$z = 1+2i$
$w = 1+i$

$$z \cdot w = (1+2i)(1-i) = 1+2 + i(2-1) = 3+i$$

• Reciproco

$x \in \mathbb{R} \setminus \{0\}$ Reciproco x^{-1} $\quad x^{-1} \cdot x = 1$

$z \in \mathbb{C} \setminus \{0\}$ Reciproco $z^{-1} = \frac{1}{z}$? NO $\quad z^{-1} \cdot z = 1 = 1 + i0$

$z = a+ib \quad a^2 + b^2 \neq 0 \iff z \neq 0$

$z^{-1} = c + id$

$\boxed{\begin{matrix} z = a+ib \\ w = c+id \end{matrix} \quad z = w \iff \begin{matrix} a=c \\ b=d \end{matrix} \quad \text{uguag. tra complessi}}$

$(a+ib)(c+id) = 1 + i0$

$ac - bd + i(bc + ad) = 1 + i0$

$\begin{cases} ac - bd = 1 \\ bc + ad = 0 \end{cases}$

Es: $z = 1+2i$
$z^{-1} = c + id$

$(1+2i)(c+id) = c - 2d + i(2c + d) = 1$

$\begin{cases} c - 2d = 1 \\ 2c + b = 0 \end{cases} \quad \begin{cases} c - 2d = 1 \\ 5c = 1 \end{cases} \quad \begin{cases} c = \frac{1}{5} \\ 2d = -1 + \frac{1}{5} \end{cases}$

Metodo più veloce per trovarlo

$\frac{1}{a+ib} \stackrel{?}{=} (a+ib)^{-1}$ NO OK

$(a+b)(a-b) = a^2 - b^2$
$(a+ib)(a-ib) =$
$a^2 + b^2 + i(-ab + ab) = a^2 + b^2$

es: $\frac{1}{3+2i} = \frac{1}{3+2i} \frac{(3-2i)}{(3-2i)} = \frac{3-2i}{9+4}$

$= \frac{3-2i}{13} = \frac{3}{13} - i\frac{2}{13}$

$\left(\frac{3}{13} - i\frac{2}{13}\right)(3+2i) = ? = 1+i0$ Si

Def

$z \in \mathbb{C}$
$z = a + ib$

CONIUGATO $\bar{z} := a - ib$

$z \cdot \bar{z} = \|z\|^2 \in \mathbb{R} \qquad z \cdot z = z^2 \in \mathbb{C}$

$\|z\| \geq 0 \to \sqrt{a^2+b^2}$

$z \cdot \bar{z} = \|z\|^2 \quad \forall z \in \mathbb{C} \qquad z = 0 + i0 \qquad \bar{z} = 0 + 0i \qquad z \cdot \bar{z} = 0 = \|0\|^2$

se $z \neq 0 + i0 \implies \|z\| \neq 0$

$\|z\| > 0$

$$\frac{z \cdot \bar{z}}{\|z\|^2} = \frac{\|z\|^2}{\|z\|^2} = 1 \qquad \text{se } z \neq 0$$

es: $z = 5 - 6i$

$$z^{-1} = \frac{5 + 6i}{25 + 36}$$

$$z \cdot \frac{\bar{z}}{\|z\|^2} = 1 \qquad \boxed{z^{-1} = \frac{\bar{z}}{\|z\|^2}}$$

$\bar{\bar{z}} = z$

$z + \bar{z} = 2\,Re(z)$ parte reale di z

$z = 5 + 12i$

$\bar{z} = 5 - 12i$

$$\mathbb{R} \Rightarrow Re(z) = \frac{z + \bar{z}}{2}$$

$$\mathbb{R} \Rightarrow Im(z) = \frac{z - \bar{z}}{2i}$$

$\to$ coeff. immaginario

Es portare nella forma

- $\dfrac{1 + i}{1 + i + 2i^2 + 3i^3 + 16} = a + ib$
- $i^{2025} = a + ib$

x casa

2) $i^{2025} = i$

1) $$\frac{1 + i}{1 + i - 2 - 3i + 16}$$

$$\frac{1 + i}{15 - 2i} \cdot \frac{15 + 2i}{15 + 2i}$$

$$\frac{15 - 2 + i(15 + 2)}{225 + 4}$$

$$\frac{13 + 17i}{229} = \frac{13}{229} + \frac{17}{229}i$$

$i^2 = -1$

$i^3 = (i^2)i = -i$

$i^4 = (-i)(i) = 1$

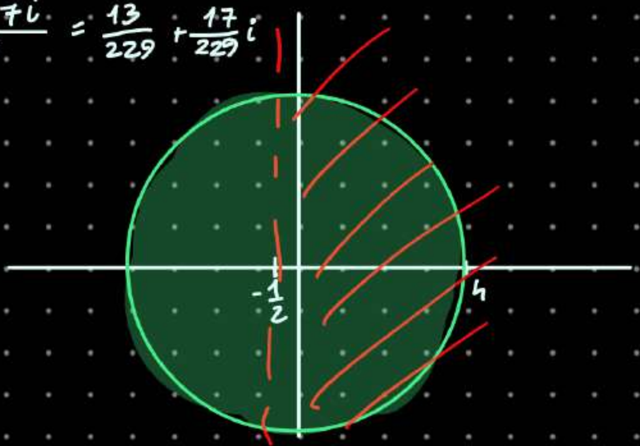
i^n

n divisibile per 2 $i = -1$

n div. per 4 $i = 1$

Es $A = \{ z \in \mathbb{C} : Re(z) > -\frac{1}{2} \}$

$= \{ z = a + ib \quad a > -\frac{1}{2} \}$



$B = \{ z \in \mathbb{C} : \|z\| \leq 4 \}$

$C = \{ z \in \mathbb{C} : \|z - 1\| > 2 \}$

$z = x + iy$

$z - 1 = x - 1 + iy$

$\|z - 1\| = \sqrt{(x-1)^2 + y^2} > 2 \to (x-1)^2 + y^2 > 4$

$= 4$

OSS $\|z\| = d(z,0)$ $\qquad$ $\|z-1\|$ è la distanza da

$1 = 1+i0$

$D = \{z : \|z-z_0\| > R\}$

TEOR Teo Fondam. dell' algebra

$z \overset{\mathbb{C}}{=} \sqrt{1+i}$

$p(z) = \sum_{k=0}^{n} a_k z^k \qquad a_n \neq 0 \qquad a_n \in \mathbb{C}$

$z \in \mathbb{C}$

$\Rightarrow \exists\, z_0 \in \mathbb{C} : p(z_0) = 0$

Polinomio z

$p(z) : z - z_0 \qquad p(z) = p_1(z)(z-z_0)$ Ruffini

grado del Polin.

$\deg P = n \qquad \deg P_1 = n-1$

$P(z) = (z-z_1)^{\mu_1} \cdot \ldots \cdot (z-z_n)^{\mu_n} \cdot$

$\mu_1 + \ldots + \mu_n = n$

Conseguenza Teo Fond.

ES $z = x + iy$ (Trovare x, y) $\in \mathbb{R}$

$z \cdot z = (x+iy)^2 = 1+i$

$x^2 - y^2 + i(xy + yx) = 1+i$

$x^2 - y^2 + 2ixy = 1+i$

$(x+iy) \neq \|x+iy\|^2$

$x^2 - 2xy + y^2 \qquad x^2+y^2$

$$\begin{cases} x^2 - y^2 = 1 \\ 2xy = 1 \end{cases}$$

$$\begin{cases} y = \frac{1}{2x} \\ x^2 - \left(\frac{1}{2x}\right)^2 = 1 \end{cases}$$

$4x^4 - 1 - 4x^2 - 1 = 0 \qquad x^2 = t$

$4t^2 - 4t - 1 = 0$

$\Delta = 32$

$t = \frac{4 \pm 4\sqrt{2}}{8} = \frac{1 \pm \sqrt{2}}{2}$

$x = \pm\sqrt{\frac{1 \pm \sqrt{2}}{2}} = \pm\sqrt{\frac{1+\sqrt{2}}{2}}$

CE $\frac{1 \pm \sqrt{2}}{2} \geq 0$ $\Rightarrow$ $\frac{1+\sqrt{2}}{2}$

$$\begin{cases} x = \pm\sqrt{\frac{1+\sqrt{2}}{2}} \\ y = \pm \frac{1}{\sqrt{\frac{1+\sqrt{2}}{2}}} \end{cases}$$

$z = a + ib$

$\frac{a}{\|z\|} = \cos(\theta)$ $\quad a = \|z\| \cos\theta$

$\frac{b}{\|z\|} = \sin(\theta)$ $\quad b = \|z\| \sin\theta$

$\underbrace{z = a + ib}_{\text{Forma algebrica}} = \overbrace{\|z\| (\cos\theta + i\sin\theta)}^{\text{F. Trigonometrica}}$

$= \underbrace{\|z\| e^{i\theta}}_{\text{F. esponenziale}}$

ES $z_1 = \rho_1 (\cos(\theta_1) + i \sin(\theta_1))$ $\quad \rho_1 \geq 0$ $\quad \theta_1$

"ro" $\quad$ Modulo $\quad$ Argomento

$\| a_1 + i b_1$ $\quad a = \rho_1 \cos\theta_1$

$b = \rho_1 \sin(\theta_1)$

$z_2 = \rho_2 (\cos\theta_2 + i \sin\theta_2)$

$z_1 \cdot z_2 = \rho_1 (\cos\theta_1 + i \sin\theta_1) \cdot \rho_2 (\cos\theta_2 + i \sin\theta_2)$

$= \rho_1\rho_2 [(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + i (\sin\theta_1 \cos\theta_2 + \cos\theta_1 + \sin\theta_2)$

$= \rho_1\rho_2 [\cos(\theta_1+\theta_2) + i \sin(\theta_1+\theta_2)]$

$= \rho (\cos\theta + i \sin\theta$

$\rho = \rho_1\rho_2$ $\quad \theta = \theta_1 + \theta_2$

$i = 1[\underbrace{\cos(\tfrac{\pi}{2})}_{=0} + i \underbrace{\sin(\tfrac{\pi}{2})}_{=1})$ $\quad \|0 + 1i\| = \sqrt{0^2+1^2} = 1$

$i^2 = i \cdot i = \underbrace{\|i\|}_{=1} \cdot \|i\| \cdot (\cos(\tfrac{\pi}{2} + \tfrac{\pi}{2}) + i \sin(\tfrac{\pi}{2} + \tfrac{\pi}{2})] = 1(\cos\pi + i \sin\pi)$

$= -1 + i0$

$z = \rho (\cos\theta + i \sin\theta)$ $\quad \rho \neq 0$

$w = \rho^{-1} (\cos(-\theta) + i \sin(-\theta))$

$z \cdot w = \rho\rho^{-1} [\cos(\theta-\theta) + i \sin(\theta-\theta)] = 1(\cos\theta + i \sin\theta)$

$= 1 + i0 = 1$

$$z^{-1} = \rho^{-1}(\cos(-\theta) + i\sin(-\theta))$$

$$= \frac{1}{\rho}(\cos(\theta) - i\sin\theta)$$

$$z^{-1} = \frac{\bar{z}}{\|z\|^2}$$

$$z = \rho(\cos\theta + i\sin\theta)$$

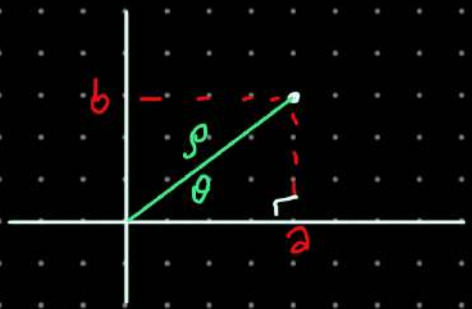
$$\frac{\bar{z}}{\rho^2} = \frac{\rho}{\rho^2}(\cos\theta - i\sin\theta) = \frac{\cos\theta - \sin\theta}{\rho}$$

$(\theta, \rho) \to (a, b)$

$a = \rho\cos\theta$

$b = \rho\sin\theta$

$(a, b) \to (\underset{\sqrt{a^2+b^2}}{\underset{\shortparallel}{\rho}}, \theta)$



$(a, b) \leftrightarrow (\rho, \theta)$

$z = a + ib$ ALG

$z = \rho(\cos\theta + i\sin\theta)$ TRIG

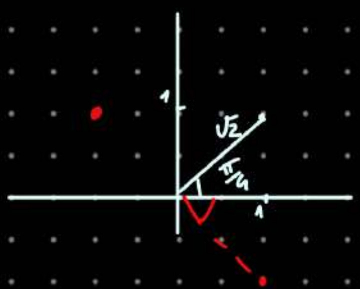
$a = \cos\theta$

$b = \sin\theta$

$\to (a, b)$ $\rho = \sqrt{a^2+b^2}$

$\tan\theta = \frac{b}{a}$ $\theta = \arctan\left(\frac{b}{a}\right)$

$z_1 = 1 + i$ $\|z_1\| \sqrt{1+1} = \sqrt{2} = \rho_1$ $\arctan\left(\frac{1}{1}\right) = \frac{\pi}{4} = \theta_1$

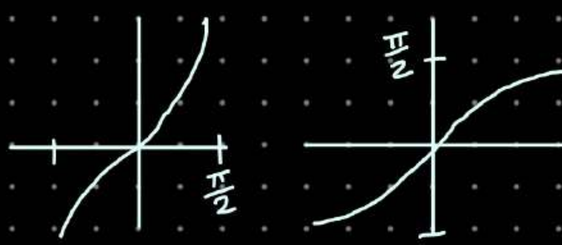


$z_2 = -1 + i$

$\rho = \sqrt{1+1} = \sqrt{2}$

$\text{arctg}\left(\frac{b}{a}\right) = \text{arctg}(1) = -\frac{\pi}{4}$

arctg é la f inversa di tg x



$$\theta \begin{cases} \text{arctg}\left(\frac{b}{a}\right) & a > 0 \\ b > 0 \quad \frac{\pi}{2} & a = 0 \\ b < 0 \quad -\frac{\pi}{2} & a = 0 \\ \text{arctg}\left(\frac{b}{a}\right) + \pi & a < 0 \end{cases}$$

$z_1 = \rho_1 (\cos\theta_1 + i\sin\theta_1)$ $\qquad$ $z_1 \cdot z_2 = \rho_1\rho_2 (\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$

$z_2 = \rho_2 (\cos\theta_2 + i\sin\theta_2)$

$z^2 = 1+i$ $\qquad$? $\qquad$ $z = \rho(\cos\theta + i\sin\theta)$

$\rho^2 = (\cos 2\theta + i\sin 2\theta) = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

$\rho^2 = \sqrt{2}$ $\qquad$ $\rho = \sqrt[4]{2} = \sqrt[2]{\sqrt{2}}$ $\qquad$ $\rho \geq 0$

- $2\theta = \frac{\pi}{4} + 2k\pi \Rightarrow \cos(2\theta) = \cos(\frac{\pi}{4} + 2k\pi)$

  $\text{sen}(2\theta) = \sin(\frac{\pi}{4} + 2k\pi)$

$\theta = \frac{\pi}{8} + k\pi \qquad k \in \mathbb{Z}$

$\sqrt{1+i}^{\,\mathbb{C}} = \{\sqrt[4]{2}[\cos(\frac{\pi}{8} + k\pi) + \sin(\frac{\pi}{8} + k\pi)$

- $\sqrt[3]{i}^{\,\mathbb{C}} = z \qquad z^3 = i \qquad z = \rho(\cos\theta + i\sin\theta)$

$z^3 = \rho^3(\cos 3\theta + i\sin 3\theta) = 1(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

$\rho = \sqrt[3]{1}^{\,\mathbb{R}} = 1$

$3\theta = \frac{\pi}{2} + 2k\pi \qquad \theta = \frac{\pi}{6} + \frac{2k}{3}\pi$

$\sqrt[3]{i}^{\,\mathbb{C}} = \{z_1, z_2, z_3\}$

$= \{\cos\frac{\pi}{6} + \frac{2k}{3}\pi + i\sin(\frac{\pi}{6} + \frac{2k}{3}\pi) \quad k \in \mathbb{Z}\}$

- $\sqrt[4]{1}^{\,\mathbb{R}} \qquad \sqrt[4]{1}^{\,\mathbb{C}} = \sqrt[4]{1+i0}^{\,\mathbb{C}}$

$z = \rho(\cos\theta + \sin\theta) \qquad \rho^4(\cos 4\theta + i\sin 4\theta) = 1(\cos\theta + i\sin\theta)$

$z^4 = 1 \qquad \rho^4 = 1 \qquad \rho = \sqrt[4]{1}^{\,\mathbb{R}}$

$4\theta = 0 + 2k\pi \qquad \theta = \frac{0}{4} + \frac{2k}{4}\pi$

$k=0 \quad \theta = 0$

$k=1 \quad \frac{2k}{4} = \frac{\pi}{2}$ $\qquad$ $k=3 \quad \frac{2\cdot 3\pi}{4} = \frac{3\pi}{2}$

$k=2 \quad \frac{2\cdot 2}{4}\pi = \pi$

$z = \rho(\cos\theta + i\sin\theta)$

$$\sqrt[n]{z} = \left\{ \rho^{1/n} \left[\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i\sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)\right], k \in \mathbb{Z} \right.$$

insieme di n elementi

Problemi con ordinam. dei $\mathbb{C}$

- $3x + 5 < 0 \qquad x \in \mathbb{R}$

  $x < -\frac{5}{3}$

- $z\,z + 5 < 0 \qquad z \in \mathbb{C}$ NON ha al cun senso

a) $i \leq 0$

$\cancel{0 + 1 \cdot i \leq 0 + i0}$

$z \leq w$

$zt \geq wt \qquad t \leq 0$

$i\,i \geq 0\,i$

$-1 \geq 0$

ASS!

b) $0 \leq i$

$0 + 0i \leq 0 + 1i$

$0\,z \leq i\,z \qquad z \geq 0$

dovrebbe rimanere uguale

$0\,i \leq i \cdot i$

$z = i$

$0 \leq -1$

Assurdo!

se avessi ordinamento

Le uniche dis. si possono fare quando si parla dei moduli

$\underset{\mathbb{R}}{\|z + 5\|} \geq \underset{\mathbb{R}}{0} \qquad z \in \mathbb{C}$

$\|z + 5\| > 0 \qquad z \in \mathbb{C} \setminus \{-5 + i0\}$

- $\boxed{|x + y| \leq |x| + |y|} \quad \boxed{\forall\, x, y \in \mathbb{R}}$

$\|z_1 + z_2\| \leq \|z_1\| + \|z_2\| \qquad \forall\, z_1, z_2 \in \mathbb{C}$

$z_1 = a_1 + ib_1$

$z_2 = a_2 + ib_2$

$\|z_1 + z_2\|^2 = (z_1 + z_2)\overline{(z_1 + z_2)} = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$

$z_1 + z_2 \quad (a_1 + a_2) + i(b_1 + b_2)$

$= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2$

$\overline{z_1 + z_2} = (a_1 + a_2) - i(b_1 + b_2)$

$= a_1 - ib_1 + a_2 - ib_2$

$= \bar{z}_1 + \bar{z}_2$

$= \|z_1\|^2 \quad + \quad \|z_2\|^2$

$\bar{\bar{z}} = z$

$\overline{z + v} = \bar{z} + \bar{v}$

$z \cdot \bar{z} = \|z\|^2$

$\overline{zw} \overset{?}{=} \bar{z} \cdot \bar{w}$

$z = \rho_z(\cos\theta_z + i\sin\theta_z)$

$w = \rho_w$ 〃 w 〃 w)

$\bar{z} = \underbrace{\rho_z(\cos\theta_z}_{a} - \underbrace{i\sin\theta_z)}_{\frac{b}{\rho}}$

$\overline{zw} = \overline{\rho\rho(\cos(\theta_z+\theta_w) + i\sin(\theta_z+\theta_w)} = \rho_z(\cos\theta_z + i\sin(-\theta_z))$

$\overline{zw} = \rho\rho(\cos(\theta_z+\theta_w) - i\sin(\theta_z+\theta_w)) = \rho_z(\cos(-\theta_z) + i\sin(-\theta_z))$

$= \rho_z(\cos\theta_z - i\sin\theta_z)\,\rho_w(\cos\theta_w - i\sin\theta_w)$

$= \bar{z}\bar{w}$

$\|z_1+z_2\|^2 = \|z_1\|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2 + \|z_2\|^2$

$\overline{z_1\bar{z}_2} = \bar{z}_1\bar{\bar{z}}_2 =$
$= \bar{z}_1 z_2$

$= \|z_1\|^2 + z_1\bar{z}_2 + \overline{z_1\bar{z}_2} + \|z_2\|^2$

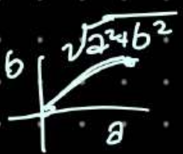
$= \underbrace{\|z_1\|^2}_{\mathbb{R}} + 2\,\text{Reale}(\underbrace{z_1\bar{z}_2}_{\mathbb{R}}) + \underbrace{\|z_2\|^2}_{\mathbb{R}}$

$\leq$

$\longrightarrow \quad \|z_1\|^2 + 2|Re(z_1\bar{z}_2)| + \|z_2\|^2$

$x \leq |x| \quad \forall x \in \mathbb{R}$
$x = 2\,Re(z_1\bar{z}_2)$

$z = a + ib$



$\to \quad |a| \leq \sqrt{a^2+b^2}$

$|Re(z)| \leq \|z\| \quad \forall z \in \mathbb{C}$ $\qquad |Im(z)| \leq \|z\|$

$\|z_1+z_2\|^2 \leq \|z_1\|^2 + 2|Re(z_1\bar{z}_2)| + \|z_2\|^2 \leq \|z_1\|^2 + 2\|z_1\bar{z}_2\| + \|z_2\|^2$

$\uparrow \qquad \uparrow$

$\|z_1 z_2\| = \|z_1\|\,\|z_2\|$

$\|z_1\bar{z}_2\| = \|z_1\| \cdot \|\bar{z}_2\|$
$= \|z_1\|\,\|z_2\|$

$z_2 = \rho_2(\cos\theta_2 + i\sin\theta_2) \quad z_1 = \rho(\cos\theta_1 + i\sin\theta_1$

$z_1 z_2 = \rho_1\rho_2(\cos(\theta_1+\theta_2) + i(\sin(\theta_1+\theta_2))$

$\sqrt{\|z_1+z_2\|^2} \leq \sqrt{\|z_1\|^2 + 2\|z_1\|\,\|z_2\| + \|z_2\|^2}$

$\|z_1+z_2\| \leq \|z_1\| + \|z_2\|$

Scrivere sotto forma di esponen

$z_1 = \rho_1(\cos\theta_1 + i\sin\theta_1) = \rho_1 e^{i\theta_1}$

$z_2 = \rho_2(\cos\theta_2 + i\,\text{sen}\,\theta_2) = \rho_2 e^{i\theta_2}$

$z_1 z_2 = \rho_1\rho_2[\cos(\theta_1+\theta_2) + i\,\text{sen}\,(\theta_1+\theta_2)]$

$\boxed{e^{i\theta} := \cos\theta + i\sin\theta \quad \forall\, \theta \in \mathbb{R}}$

$\rho_1 e^{i\theta_1} \cdot \rho_2 e^{i\theta_2} =$
$= \rho_1\rho_2\, e^{i(\theta_1+\theta_2)}$

$e^z \qquad z = a+ib$

$e^z = e^{a+ib} = e^a \cdot e^{ib} := e^a[\cos b + i\sin b]$

$: e^z \cdot e^w = e^{z+w}$

$e^x = y \qquad \overset{x,y\in\mathbb{R}}{x = \ln y} \quad y>0 \qquad e^{\ln y} = y \qquad \forall y>0$

Continuità di una F

$f: D \to \mathbb{R}$, $D \subseteq \mathbb{R}$ È continua in $x_0 \in D$

$(f(x) = \frac{1}{x} = 0 \quad x \neq 0 \quad f$ è continua in x_0 ?

DEF $\{x_n\} \in D$ $\lim_{n\to+\infty} x_n \to x_0 \Rightarrow \lim_{n\to+\infty} f_n = f(x_0)$ $\quad f_n := f(x_n)$

PER OGNI

ES $\quad x_n = \frac{1}{n} \to x_0 = 0 \qquad g(x_n) = \frac{1}{x_n} = \frac{1}{\frac{1}{x_n}} = n \to +\infty$

$y_n \equiv 0 \quad \forall n \in \mathbb{N}$

$y_n \to 0 \qquad g(y_n) = 5 \quad \forall \ldots$

DEF $\quad f$ è continua in $A \subseteq D$

$\boxed{f \in C(A)}$

DEF $\quad f$ è continua in tutti i punti $x \in A$

DEF $\quad f: D \to \mathbb{R} \quad x_n \in D \quad f$ è continua in $x_0 \in D$

$\forall \varepsilon > 0 \ \exists\, \delta > 0$ t.c. $x \in D \ |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$

Prop: f, g continue in x_0, $f+g$, $f \cdot g$, $\frac{f}{g}$ $g \neq 0$ // sempre continue

---

$f(x) = x$ è continua $\forall x \in \mathbb{R}$ $f(x) = x \in C(\mathbb{R})$

$x \cdot x$ " "

$$P(x) = \sum_{n=0}^{M} a_n x^n$$

$$\lim_{n \to +\infty} \sin\left(\frac{1}{x}\right) = 0 \qquad x_n = \frac{1}{n} \to 0$$

$$f\left(\frac{1}{n}\right) = \sin\left(\frac{1}{n}\right) \to 0 \quad = \sin(0) = f(x_0)$$

$0 \leq |\sin(x)| \leq |x|$ $\qquad x_n \to 0$ qualsiasi $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$\underset{\downarrow}{0} \leq \underset{\downarrow}{|\sin}(x_n)| \leq \underset{\downarrow}{|x_n|}$$
$$0 \qquad\quad 0 \qquad\qquad 0$$

- $\sin(p) - \sin(q) = 2\cos\left(\frac{p+q}{2}\right) \sin\left(\frac{p-q}{2}\right)$ $\quad \forall p, q \in \mathbb{R}$
  Form. PROSTAFERESI

$f(x) = \sin(x)$ $\downarrow$ continua in $x_0 \in \mathbb{R}$

$\forall \varepsilon > 0 \ \exists \delta > 0$ t.c. $\forall x \in \mathbb{R}$ $|x - x_0| < \delta$ $|f(x) - f(x_0)| < \varepsilon$

$\forall \varepsilon > 0 \ \exists \delta > 0$ t.c. $\forall x \in \mathbb{R}$ $|x - x_0| < \delta$ $|\sin x - \sin x_0| < \varepsilon$

$$|\sin x - \sin x_0| = \left|2\cos\left(\frac{x+x_0}{2}\right) \sin\left(\frac{x-x_0}{2}\right)\right| \leq 2\left|\cos\frac{x+x_0}{2}\right| \left|\sin\frac{x-x_0}{2}\right|$$

$\frac{|x - x_0|}{2} < \frac{\pi}{2}$

$0 < \delta < \pi$

$$2 \cdot 1 \left|\frac{x-x_0}{2}\right| = \cancel{2} \cdot 1 \left|\frac{x-x_0}{\cancel{2}}\right| = |x - x_0| < \varepsilon$$

TEOREMA Weierstrass

$$f \in C([a, b]) \implies \exists\, m, M \in \mathbb{R} \qquad m = \min_{[a,b]} f \quad M = \max_{[a,b]} f$$

No DIMOSTRAZ.

ES: Trovare $f : (0, 1] \to \mathbb{R}$ continua in tutti i punti

dove $\max(F)$ N.E.
$\min(F)$ N.E.

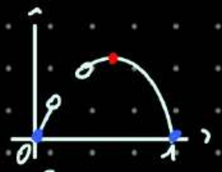
TEOREMA di Weierstrass

[Vaierstrass] anni 800

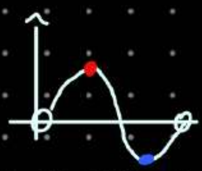
ESISTENZA

Sia $[a,b] \subseteq \mathbb{R}$ un intervallo (chiuso e limitato)
e $f:[a,b] \to \mathbb{R}$
se f è continua su $[a,b]$ (estremi inclusi) allora
esistono il max e min glob.
di f su $[a,b]$

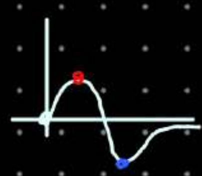
Commenti ① le hp sono SUFFICIENTI ma NON necessari



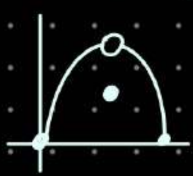
f non cont.



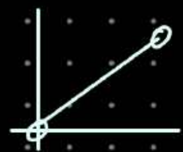
int Aperto



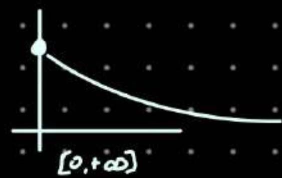
② Per avere la Tesi servono TUTTE E TRE le ipotesi



Non c'é MAX



Né min
Né MAX



Non c'é min

Teorema permanenza segno

f continua in $x_0 \in D$ $f(x_0)>0$

$\Rightarrow \exists\, \delta>0$ T.C. $\forall x \in D$ $|x-x_0|<\delta$

(No perdita generalità) $f(x)>0$

DIM $\forall \varepsilon>0$ $\exists\, \delta>0 :$ $x \in D$ $|x-x_0|<\delta$ $|f(x)-f(x_0)|<\varepsilon$

$$-\varepsilon < f(x)-f(x_0) < \varepsilon$$

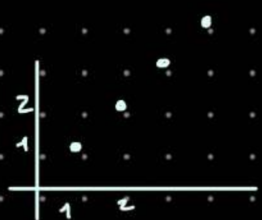
$$0< \frac{f(x_0)}{2} = f(x_0)-\varepsilon < f(x) < f(x_0)+\varepsilon$$

$\varepsilon = \frac{f(x_0)}{2} > 0$

$f(x_0)-\varepsilon = f(x_0) - \frac{f(x_0)}{2}$ $= \frac{f(x_0)}{2} > 0$

Es $f(x)=n$ $x=n$

$D=\{1,2,3,\dots\}$



è continua in $x_0=2$

$\forall \varepsilon>0$ $\exists\, \delta>0 :$ $x \in D$ $|x-2|<\delta$ $|f(x)-f(2)|<\varepsilon$

se $\delta=\frac{1}{2}$ $x \in (2-\frac{1}{2}, 2+\frac{1}{2}) \Leftrightarrow x=2 \Rightarrow |f(2)-f(2)|=0<\varepsilon$

f è continua da DESTRA in x_0

$x_n \to x_0$ $\quad x_n \in D \quad x_n \geq x_0$

$$\lim_{n \to +\infty} f(x_n) = f(x_0)$$

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ T.C. } \forall x \in D \ \underset{x > x_0}{(x - x_0) < \delta} \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

Teorema DI ESISTENZA DEGLI ZERI

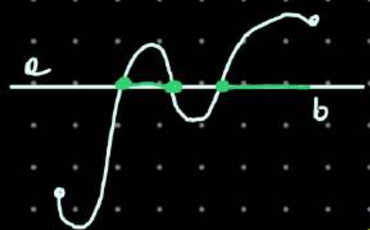
Sia $F: [a, b] \to \mathbb{R}$ e supponiamo che
1. F sia continua su $[a, b]$
2. $F(a)$ e $F(b)$ sono discordi (ossia $F(a) \cdot F(b) < 0$)

$\Rightarrow$ Allora esiste ALMENO un $c \in (a, b)$ tale che $F(x) = 0$

in $x = c$ il grafico interseca asse x

<u>OSS</u> Si tratta di un teorema di esistenza, quindi NON DICE né quanti né dove sono i punti di intersezione con asse x

<u>DIM</u> Supponiamo che $F(a) < 0$ e $F(b) > 0$ (senza perdere generalità $\Rightarrow F(a) > 0$ e $F(b) < 0$)



$$c = \inf \{x \in [a, b] : F(x) \geq 0\}$$

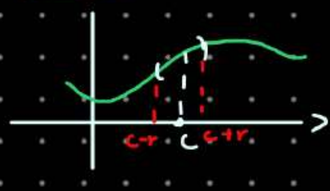
esiste sempre tranne per $\emptyset$

questo insieme è $\neq \emptyset$ perché b vi appartiene per ipotesi

$\Rightarrow$ L'inf ESISTE

1. Supp per assurdo che $F(c) > 0$ dato che F è continua $F(c) = \lim_{x \to c} F(x) > 0$, quindi per il Teor. della perm. del segno, esiste r tale che $F(x) > 0$ se $x \in (x - r, x + r)$

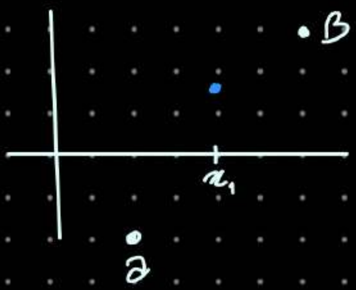
Questo va CONTRO la def di c con inf degli x tali che $F(x) \geq 0$



2. analogo $F(c) < 0$

DIMOST.
metodo Berselli
Metodo Bisezione

$f(a)<0, \quad f(b)>0$



$x_1 = \frac{a+b}{2}$ (P.to medio

$f(x_1)$ può avere tre casi

① $f(x_1)=0$ ✓

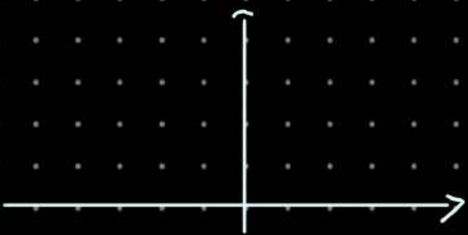
② $f(x_1)<0$
$f\left[\frac{a+b}{2}, b\right]$
$f\left(\frac{a+b}{2}\right)<0$
$f(b)>0$

③ $f(x_1)>0$
$f\left[a, \frac{a+b}{2}\right]$
$f(a)<0$
$f\left(\frac{a+b}{2}\right)>0$

$$I_1 = [a_1, b_1] = [a, b]$$

$$f\left(\frac{a_1+b_1}{2}\right) = \begin{cases} >0 & \left[a_1, \frac{a_1+b_1}{2}\right] = I_2 = [a_2, b_2] \\ =0 & \text{STOP} \\ <0 & \left[\frac{a_1+b_1}{2}, b_1\right] = I_2 = [a_2, b_2] \end{cases}$$

$\{a_n\}$ $\quad a_n \to c \in [a,b] \qquad f(c)=0$
$\{b_n\}$ $\quad b_n \to c$



$f(x) = x^3 + x - 1$ • Trovare $x : f(x)=0$

$f_n = f(n) = n^3 + n - 1 \qquad \lim_{n \to +\infty} f_n = +\infty$

$\Rightarrow N_1 \in \mathbb{N} : f_n > 0 \quad \forall n > N_1$

$g_n = f(-n) = (-n)^3 - n - 1 \qquad \lim_{n \to +\infty} g_n = -\infty$

$\Rightarrow \exists N_2 \in \mathbb{N} \qquad g_n < 0 \quad \forall n > N_2$

$\Rightarrow I[-n, n] \qquad f(-n)<0 \qquad f(n)>0 \qquad f \in C([-n,n])$

$f(0) = -1$

$f(1) = 1 + 1 - 1$

$ax^2+bx+c=0$ $\qquad$ $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$

$z^3+az^2+bz+c=0$ $\qquad$ $z=x-\dfrac{a}{2}$

$z^3=\left(x-\frac{a}{3}\right)^3=x^3+3x^2\left(-\frac{a}{3}\right)+3x\left(-\frac{a}{3}\right)^2-\left(-\frac{a}{3}\right)^2$

$x^3-b'x+c'=0$ $\qquad$ Cubica Depressa 
(Dal termine in grado 2)

$x^3=mx+n$ $\qquad$ $$x=\sqrt[3]{\frac{n}{2}+\sqrt{\frac{n^2}{4}-\frac{m^3}{87}}}+\sqrt[3]{\frac{n}{2}-\sqrt{\frac{n^2}{4}-\frac{m^3}{87}}}$$

$x^3=6x+4$ $\qquad$ $\sqrt[3]{2+2i}+\sqrt[3]{2-2i}$

$$\boxed{\begin{aligned}x_1&=-2\\ x_{\dots}&=1-\sqrt{2}\\ x_{\dots}&=1+\sqrt{3}\end{aligned}}$$

$x=\sqrt[3]{p}+\sqrt[3]{q}$

$x^3=(\sqrt[3]{p}+\sqrt[3]{q}$

$x^3=(\sqrt[3]{p}+\sqrt[3]{q})^3=p+3\sqrt[3]{p^2q}+\sqrt[3]{pq^3}+q=p+3\sqrt[3]{}$

$x^3=3\sqrt[3]{pq}\,x+p+q$ $\qquad$ $\begin{cases}p+q=n\\ 3\sqrt[3]{pq}=m\end{cases}$

$27pq=m^3$

$\begin{cases}pq=\dfrac{m^3}{27}\\ p+q=n\end{cases}$ $\qquad$ $\begin{cases}4pq=4\dfrac{m^3}{27}\\ p^2+2pq+q^2=n^2\end{cases}$

$\begin{cases}p-q=\sqrt{n^2-\frac{4}{27}m^3}\\ p+q=n\end{cases}$

$\begin{cases}p=\left(n+\sqrt{n^2-\frac{4}{27}m^3}\right)\dfrac{1}{2}\\ q=\left(n-\sqrt{n^2-\frac{4}{27}m^3}\right)\dfrac{1}{2}\end{cases}$

ES $\quad$ Calcolare $\quad i^i=e^{i\ln i}$ $\qquad$ (famiglia di Reali)

Teorema dei valori intermedi (Darboux)

Sia f una funzione continua su $[a,b]$

$\exists \quad m = \min_{[a,b]} f \quad e \quad M = \max_{[a,b]} f \quad$ per weierstrass

$\forall \lambda \in [m, M] \quad \exists \, x_0 \in [a,b] \quad t.c. \quad f(x_0) = \lambda$

la funzione continua assume tutti i valori che vanno dal minimo al massimo

OSS la funzione deve essere continua con il dominio senza buchi

DIM

consideriamo

$\lambda > m \quad e \quad \lambda < M$

$f(x_m) = m \qquad f(x_M) = M$

$m < \lambda < M$

$F := f(x) - \lambda \qquad x_0 \text{ è t.c. } f(x_0) = \lambda \iff F(x_0) = 0$

$f \in C([a,b]) \implies F \in C([a,b])$

$F(x_m) = f(x_m) - \lambda = m - \lambda < 0$

$F(x_M) = f(x_M) - \lambda = M - \lambda > 0$

$F|_{[x_m, x_M]} \in C([x_m, x_M]) \qquad F|_{[x_m, x_M}$

$\implies$ Teo zeri applicato a F assicura $\exists \, x_0 \in$

TEOREMA $f \in C([a,b])$ invertibile $(\exists\, g = f^{-1})$

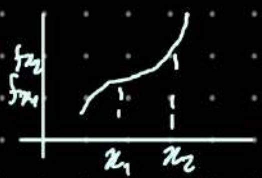
$f: [a,b] \to [c,d]$ $\qquad$ $g: [c,d] \to [a,b]$

T.C. $g(f(x)) = x$ $\qquad$ $f(g(y)) = y$

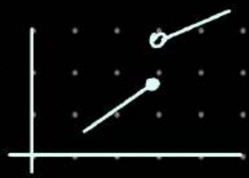
$\forall x \in [a,b]$ $\qquad$ $\forall y \in [c,d]$

Allora f é monotona in senso stretto

OSS



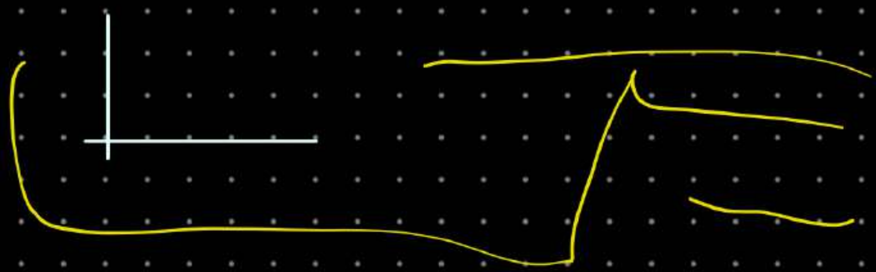
Strett cresc



Non continua

Strett. cresc. $\Rightarrow$ invertibile

Dim



Teo $f \in C([a,b])$ invertibile $\Rightarrow g = f^{-1}: [c,d] \to [a,b]$

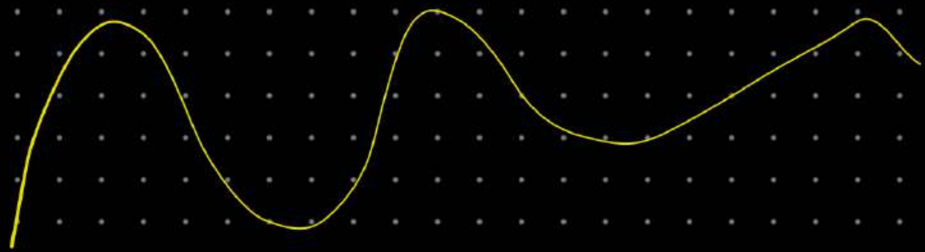
$f: [a,b] \to [c,d] \Rightarrow g \in C([c,d])$

TEOREMA $f_1: D_1 \to \mathbb{R}$ $\quad f_2: D_2 \to \mathbb{R}$ $\quad D_2 \subseteq Im(f_1)$

$f_2(f_1(x)) \in$

$f(x) = x^2$ $\quad [a,b] = [1,7]$

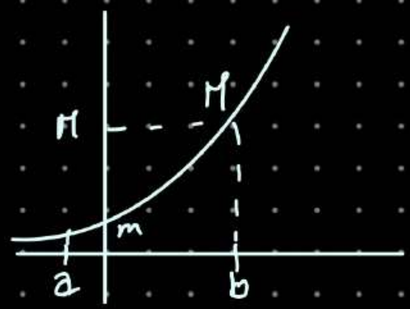
$f: [1,7] \to [1,49]$ $\qquad g(y) = \sqrt{x} : [1,49] \to [1,7]$



ES $f(x) = e^x$ $\quad f: \mathbb{R} \to (0, +\infty)$

$g(y) = \ln(y)$

$g(f(x)) = x, \quad f(g(y)) = y \quad \begin{array}{l} \forall x \in \mathbb{R} \\ \forall y > 0 \end{array}$



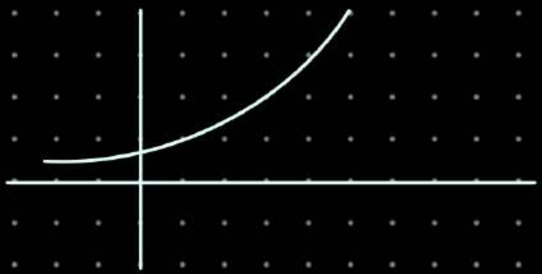
$\operatorname{Im} f(x) = (m, M)$

$f(x) = e^x_{|[a,b]}$

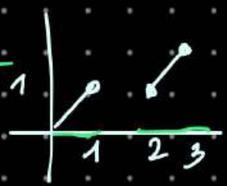
$m = f(a) = m \qquad m = f(b) = M$

$e^x: [a,b] \to [e^a, e^b]$

$\ln y: [e^a, e^b] \to [a, b]$



ES



$f(x) \begin{cases} x & x \in [0,1] \\ x-1 & x \in [2,3] \end{cases}$

$D = [0,1] \cup [2,3]$

$f \in C(D) = C([0,1] \cup [2,3])$

$\operatorname{Im}(f) = [0,2]$

$0 \le y \le 1 \qquad x = y \qquad f(x)$

$1 < y \le 2 \qquad \boxed{y = x-1} \qquad y = y+1$

$g(y) \begin{cases} y & y \in [0,1] \\ y+1 & y \in (1,2) \end{cases} \qquad \cup [0,2] \to [\quad]$

l'inversa non è invertibile in quanto il dom di partenza non è un intervallo singolo

ES $f(x) = \tan x \qquad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f \in C((a,b)) \not\Rightarrow$ $m = \min f$, $M = \max f$ NON ESISTONO

TEOREMA VALORI INTERMEDI (DLC)

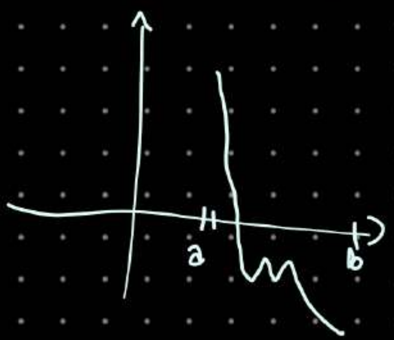
Sia f continua sull' intervallo $(-a,b)$

$\forall \lambda \in \left(\inf\limits_{(a,b)} f, \sup\limits_{(a,b)} f\right)$

$\Rightarrow \exists\, x_0 \in (a,b)$ t.c. $f(x_0) = \lambda$

Dim $f : (a,b) \to \mathbb{R} \quad f \in C((a,b))$

$-\infty < \inf\limits_{(a,b)} f \qquad \sup\limits_{(a,b)} f = +\infty \qquad \forall \lambda \in \left(\inf\limits_{(a,b)} f, +\infty\right)$



$i = \inf\limits_{(a,b)} f$

① $i \leq f(x)$ (i é un minorante)

② $\forall \varepsilon > 0 \ \exists\, \bar{x} \in (a,b)$ [i é il max dei minoranti]
$f(\bar{x}) < i + \varepsilon$

$i < \lambda \qquad \lambda$ e' strett. magg dell' $\inf\limits_{(a,b)} f$

$\Rightarrow \exists\, \varepsilon > 0$ t.c. $i + \varepsilon < \lambda$

uso la def di inf $\Rightarrow \exists\, \bar{x} \in (a,b) : f(\bar{x}) < i + \varepsilon < \lambda$

$\sup\limits_{(a,b)} f = +\infty \iff f$ non e' limit sup. $\iff \forall M \in \mathbb{R}$
$\exists\, \bar{\bar{x}} \in (a,b)$ t.c.
$f(\bar{\bar{x}}) > M$

$M = \lambda \quad \exists\, \bar{\bar{x}} \in (a,b)$ t.c. $f(\bar{\bar{x}}) > \lambda$

$\bar{x}, \bar{\bar{x}} \in (a,b) \quad \begin{cases} f(\bar{x}) < \lambda \\ f(\bar{\bar{x}}) > \lambda \end{cases} \Rightarrow$ Deve toccare λ almeno una volta

$I = [\bar{x}, \bar{\bar{x}}]$ se $\bar{x} < \bar{\bar{x}}$
$I = [\bar{\bar{x}}, \bar{x}]$ se $\bar{\bar{x}} < \bar{x}$